

EE104 Fundamentals of Electric Circuits (Lesson 10)

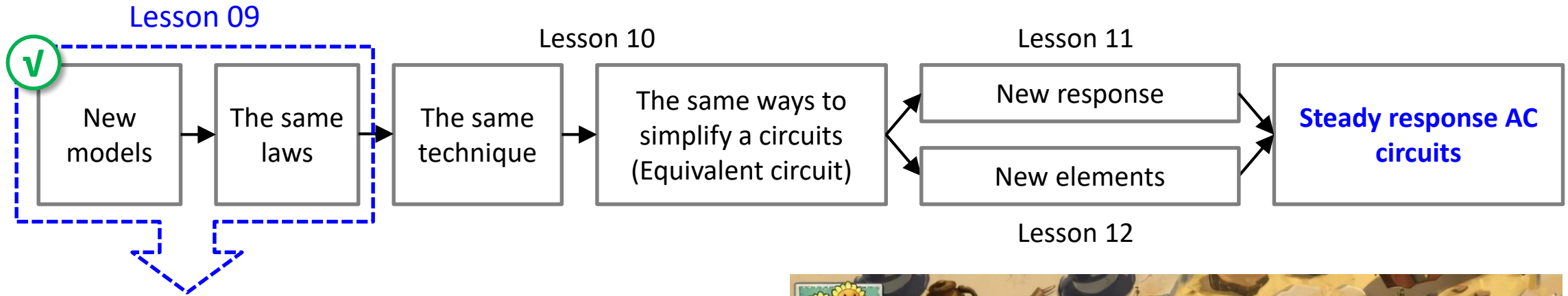
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Review of Lessons 9(a)

- EE104 road map



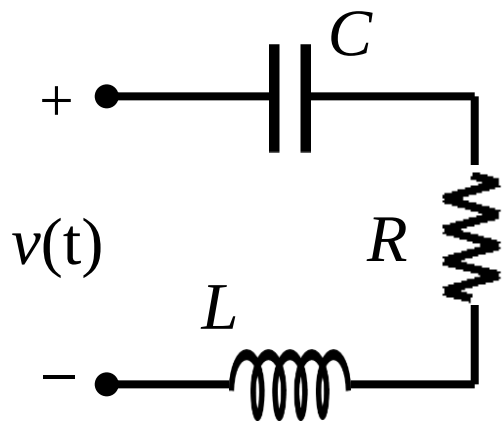
- Sinusoids and phasors
- AC circuits in frequency domain
- Basic law
- Phase shifters



Review of Lessons 9(b)

Phasors: $v(t) = \text{Re}(V_m e^{j\phi} e^{j\omega t}) = \text{Re}(\mathbf{V} e^{j\omega t})$

(a) Simplified analysis and calculation:



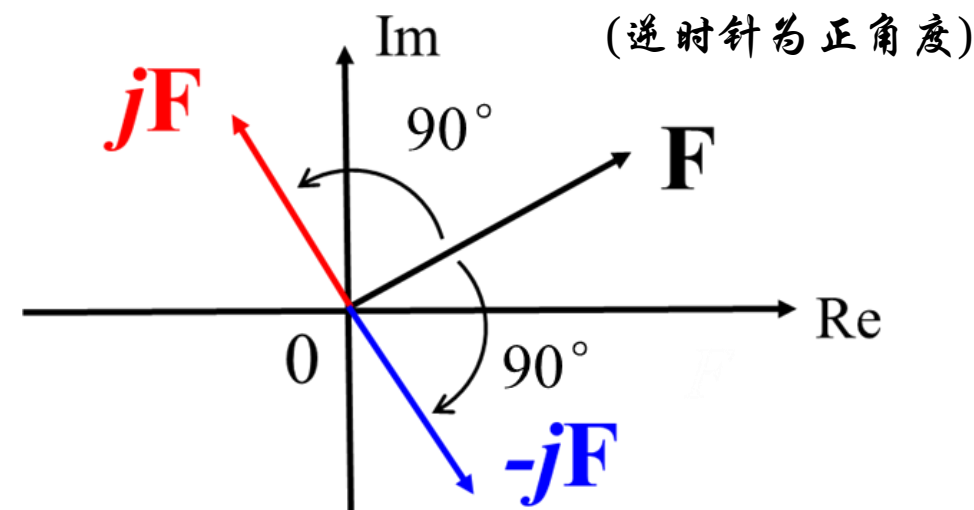
$$i(t) = \sqrt{2} I \cos(\omega t + \psi_i)$$

$$u(t) = Ri + L \frac{di}{dt} + \frac{1}{C} \int i dt$$



$$\mathbf{V} = R\mathbf{I} + j\omega L\mathbf{I} + \frac{\mathbf{I}}{j\omega C}$$

*j is a rotating factor:



$$e^{\pm 90^\circ} = \cos 90^\circ + j \sin 90^\circ = \pm j$$

Review of Lessons 9(c)

Phasors: $v(t) = \text{Re}(V_m e^{j\phi} e^{j\omega t}) = \text{Re}(\mathbf{V} e^{j\omega t})$

(b) Thanks to phasors, Ohm's Law can be used for AC circuits with the same ω :

Resistor	Inductor	Capacitor
$\mathbf{V} = R\mathbf{I},$	$\mathbf{V} = j\omega L\mathbf{I},$	$\mathbf{V} = \frac{\mathbf{I}}{j\omega C}$
$\frac{\mathbf{V}}{\mathbf{I}} = R,$	$\frac{\mathbf{V}}{\mathbf{I}} = j\omega L,$	$\frac{\mathbf{V}}{\mathbf{I}} = \frac{1}{j\omega C}$

$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} \quad \text{or} \quad \mathbf{V} = \mathbf{Z}\mathbf{I}$

$$\mathbf{Z} = R + jX = |\mathbf{Z}| \angle \theta$$

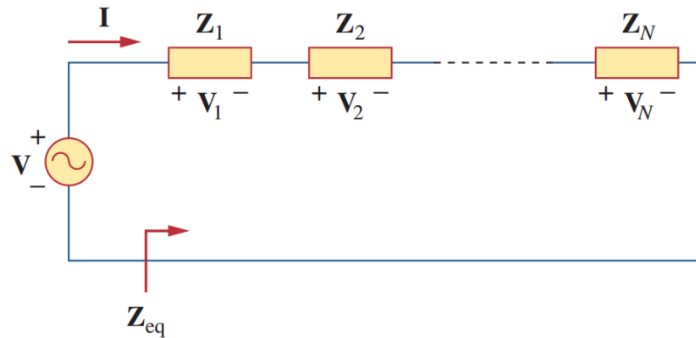
	$\mathbf{Z}$	R	X
R	R	R	-
L	$j\omega L$	-	ωL
C	$1/j\omega C$ $= -j/\omega C$	-	$-1/\omega C$

Review of Lessons 9(d)

Phasors: $v(t) = \text{Re}(V_m e^{j\phi} e^{j\omega t}) = \text{Re}(\mathbf{V} e^{j\omega t})$

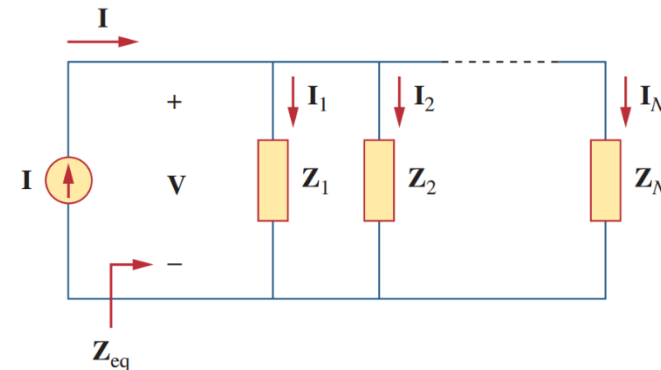
(c) Thanks to the phasors, KCL and KVL can be used for AC circuits with the same ω :

$$\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_n = 0$$



$$\mathbf{Z}_{eq} = \frac{\mathbf{V}}{\mathbf{I}} = \mathbf{Z}_1 + \mathbf{Z}_2 + \cdots + \mathbf{Z}_N$$

$$\mathbf{I}_1 + \mathbf{I}_2 + \cdots + \mathbf{I}_n = 0$$



$$\frac{1}{\mathbf{Z}_{eq}} = \frac{\mathbf{I}}{\mathbf{V}} = \frac{1}{\mathbf{Z}_1} + \frac{1}{\mathbf{Z}_2} + \cdots + \frac{1}{\mathbf{Z}_N}$$

Review of Lessons 9(e)

Impedance: $\mathbf{Z} = R + jX = |\mathbf{Z}| \angle \theta$

It is important to understand the impedance:

$$|\mathbf{Z}| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1} \frac{X}{R}$$

$$R = |\mathbf{Z}| \cos \theta, \quad X = |\mathbf{Z}| \sin \theta$$

- If X is positive, inductive impedance (感性阻抗或滞后阻抗), i lags v .
- If X is negative, capacitive impedance (容性阻抗或超前阻抗), i leads v .

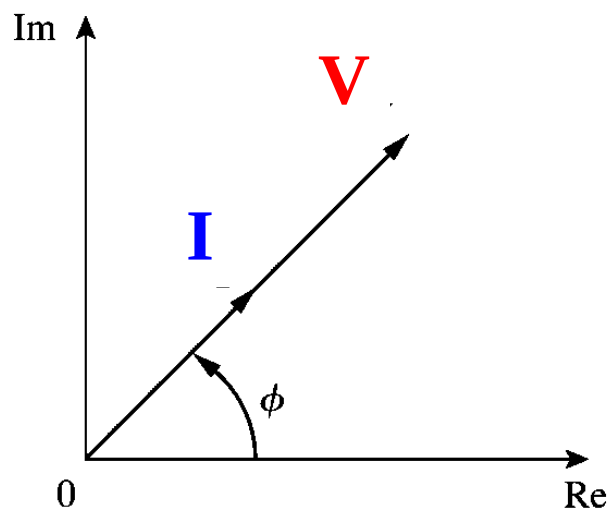
$$\begin{aligned} \mathbf{V} &= \mathbf{Z}\mathbf{I} = |\mathbf{Z}| \angle \theta \times |\mathbf{I}| \angle \phi \\ &= |\mathbf{Z}| \angle \theta \times |\mathbf{I}| \angle \phi \\ &= |\mathbf{Z}| |\mathbf{I}| \angle (\phi + \theta) \\ &= \sqrt{R^2 + X^2} |\mathbf{I}| \angle \left(\phi + \tan^{-1} \frac{X}{R} \right) \end{aligned}$$

An impedance can modify both the amplitude as well as the phase.

Review of Lessons 9(f)

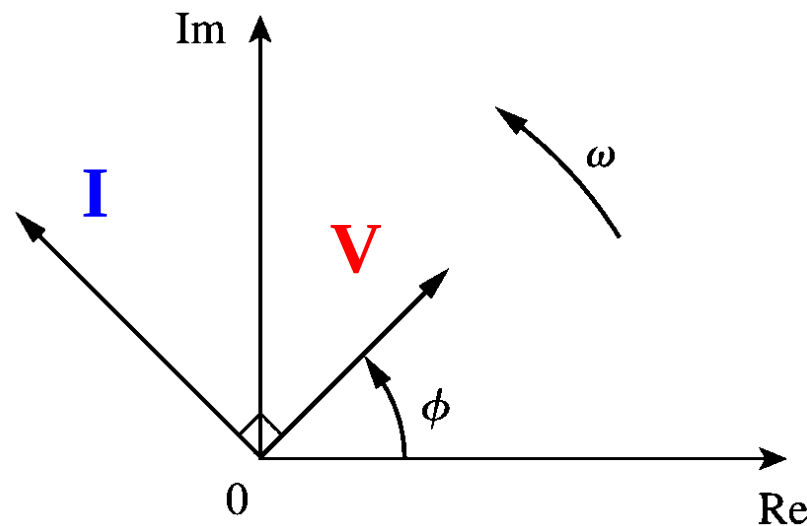
CIVIL

Resistors



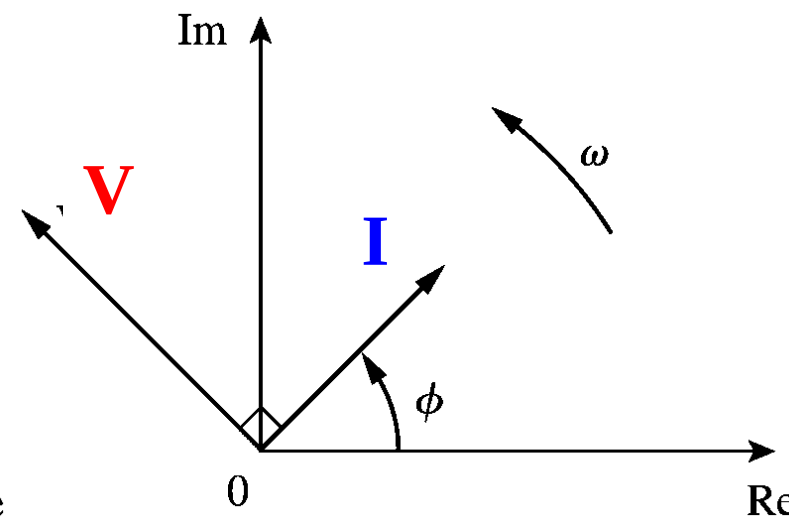
I and V in phase

Capacitors



I leads V

Inductors

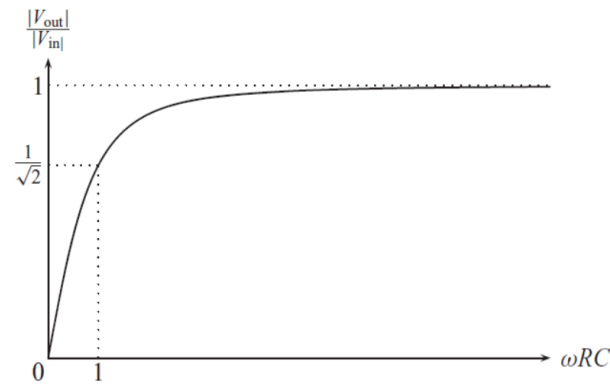
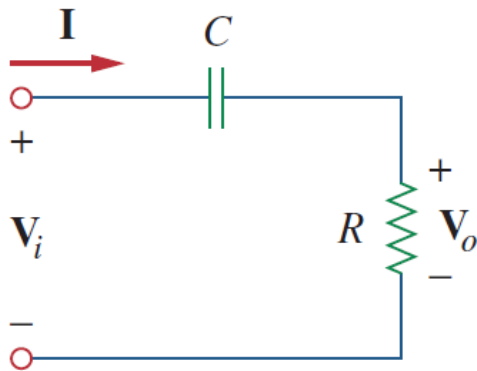


I lags V

Review of Lessons 9(g)

Phase shifters / voltage dividers

Positive

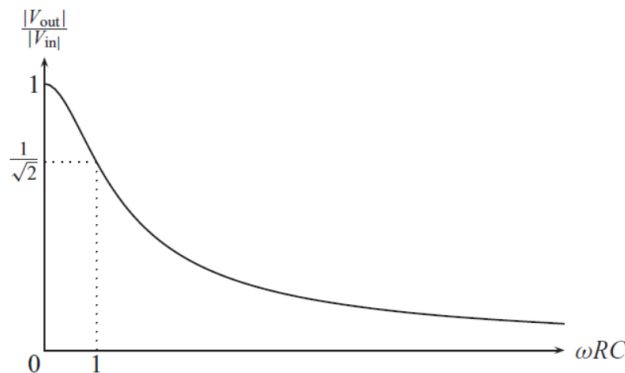
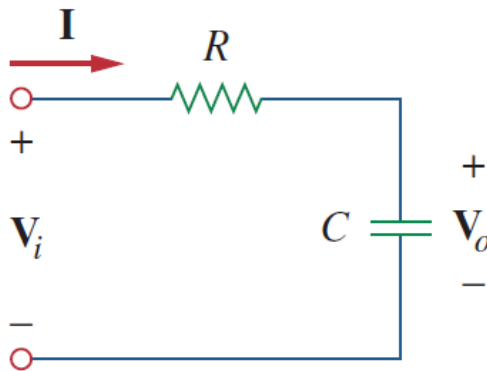


$$\frac{|V_o|}{|V_i|} = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}}$$

$$\theta = \tan^{-1}\left(\frac{1}{\omega RC}\right)$$

$$\theta \rightarrow \frac{\pi}{2}$$

Negative

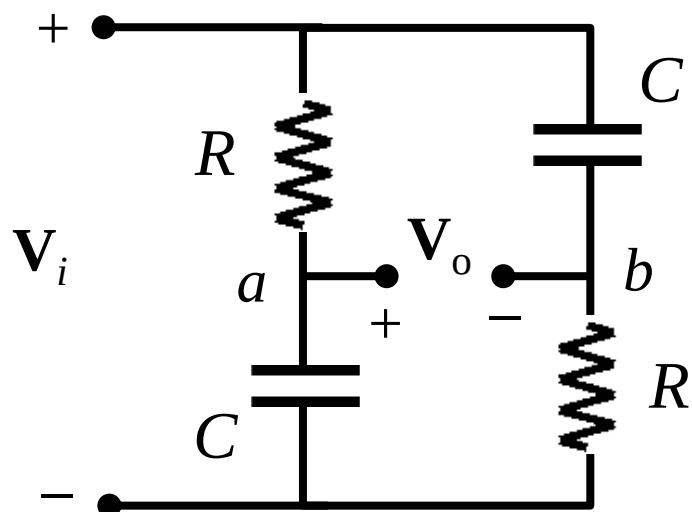


$$\frac{|V_o|}{|V_i|} = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$\theta = -\tan^{-1}(\omega RC)$$

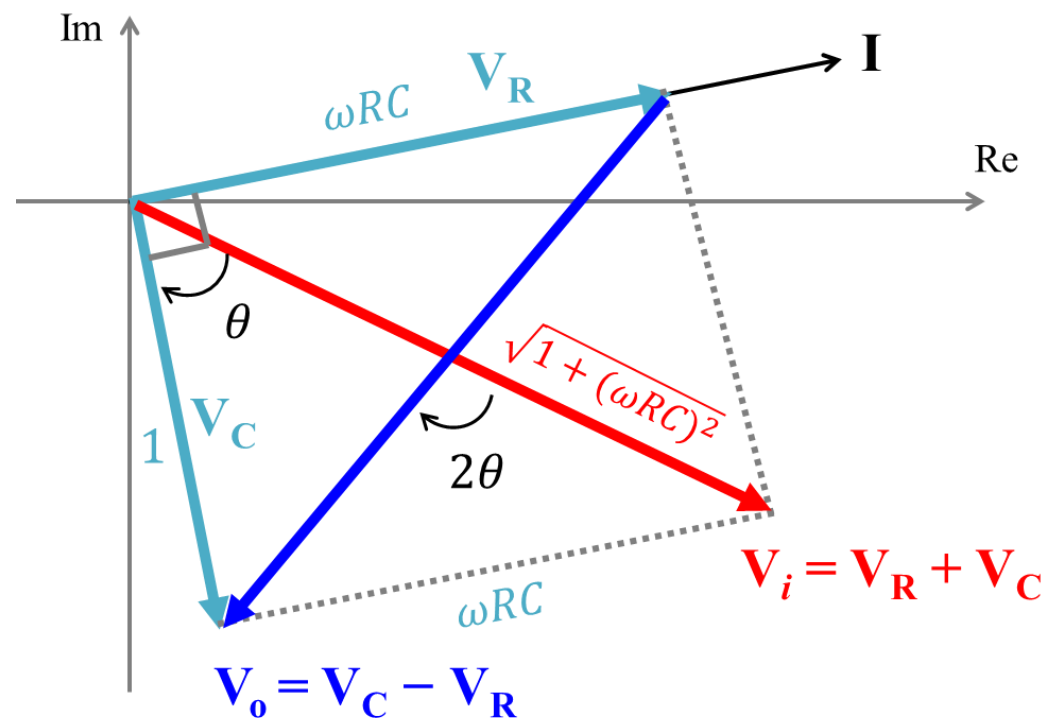
$$\frac{|V_o|}{|V_i|} \rightarrow 0$$

Phase shifters / AC voltage dividers



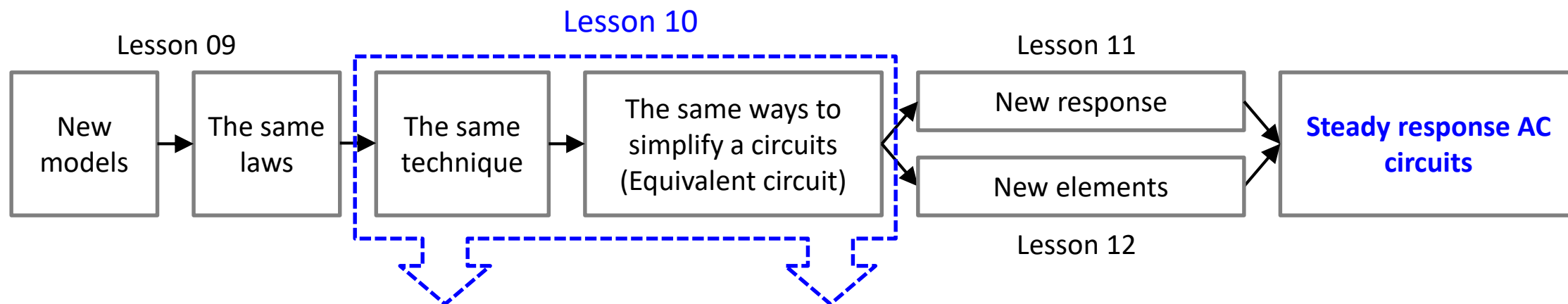
When $V_o = V_a - V_b$

$$|V_o| = |V_i|, \quad \theta = -2\tan^{-1}(\omega RC)$$



Abstract of Lesson 10 (a)

- EE104 road map



Techniques and theorems

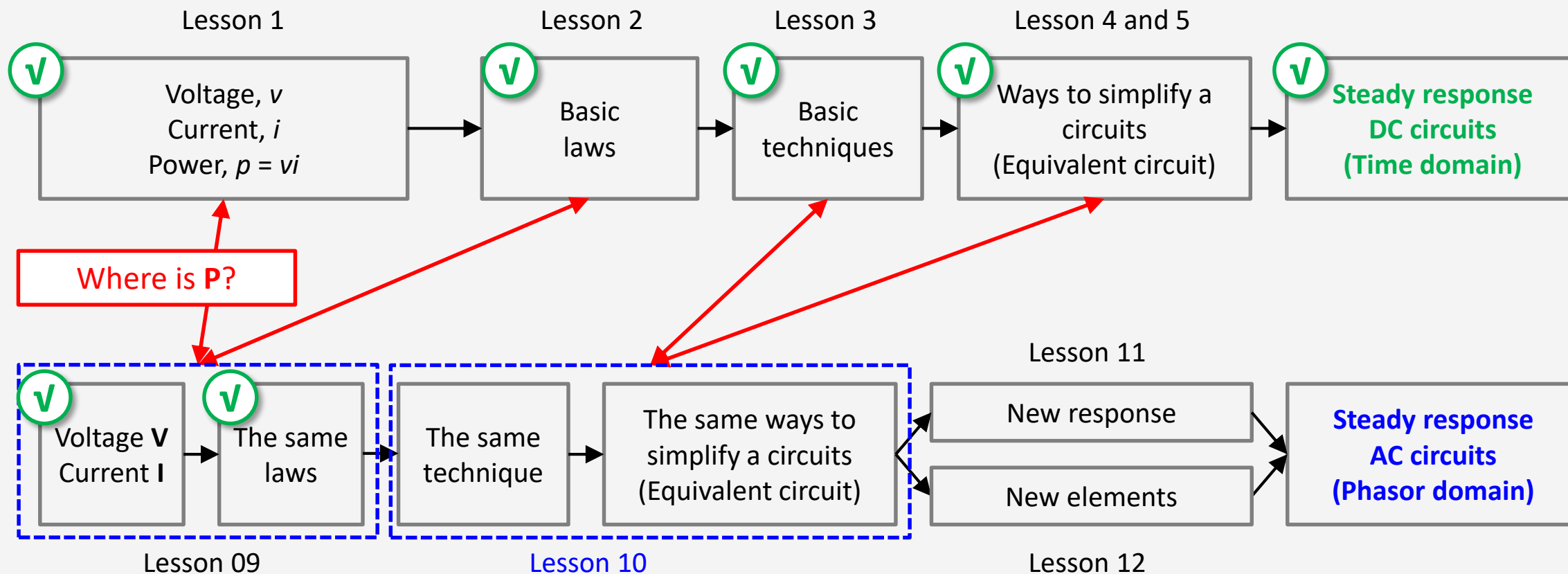
- Nodal analysis and mesh analysis
- Superposition and source transformation
- Thevenin and Norton theorems
- AC op amp circuits

Power

- Power in time domain
- Power in phasor domain (complex power)
- Power factor
- Maximum power transfer theorem

Abstract of Lesson 10 (b)

- EE104 road map: comparison of steady-state AC and DC circuits roadmaps

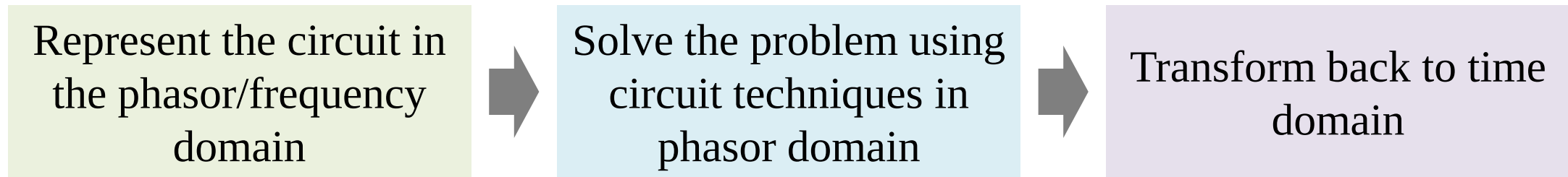


Lesson 11 Steady-state and power analysis of AC circuits

10.1	Nodal and mesh analyses	Slides 13-24
10.2-10.3	Superposition theorem and source transformation	Slides 26-40
10.4	Thevenin and Norton equivalent circuits	Slides 42-52
10.5	AC circuits with op amp	Slides 54-58
10.6	AC power in time domain	Slides 60-71
10.7	AC power in phasor domain	Slides 73-83
10.8	Summary and assignment	Slides 85-86

10.1 Nodal and mesh analyses (a)

- How to analyze a steady-state AC circuit:



- a) Making use the generalized Ohm's law, the procedure will be simpler than direct analysis in time domain.
- b) As you will see later, the techniques we've learned for DC circuits analysis can be equally applied.
- c) Noted that phasors can be used only when the circuit has one frequency.

10.1 Nodal and mesh analyses (b)

- Recall: nodal analysis

Nodal voltage analysis is a circuit analysis technique, in which nodal voltages are selected as the independent variables. Nodal analysis is based on Kirchhoff's Current Law (KCL) with coordination of Ohm's law.

Step 01

Label the Nodes

- ❑ Select a reference node.
- ❑ Prefer ground as the reference
- ❑ Name voltages at each node.

Step 02

KCL at nodes

- ❑ Apply KCL at each node except the reference node
- ❑ Write down the equation at each node

Step 03

Ohm's Law

- ❑ Apply Ohm's Law to express each current
- ❑ Generally the i flows from higher to lower potential

Step 04

Solve equations

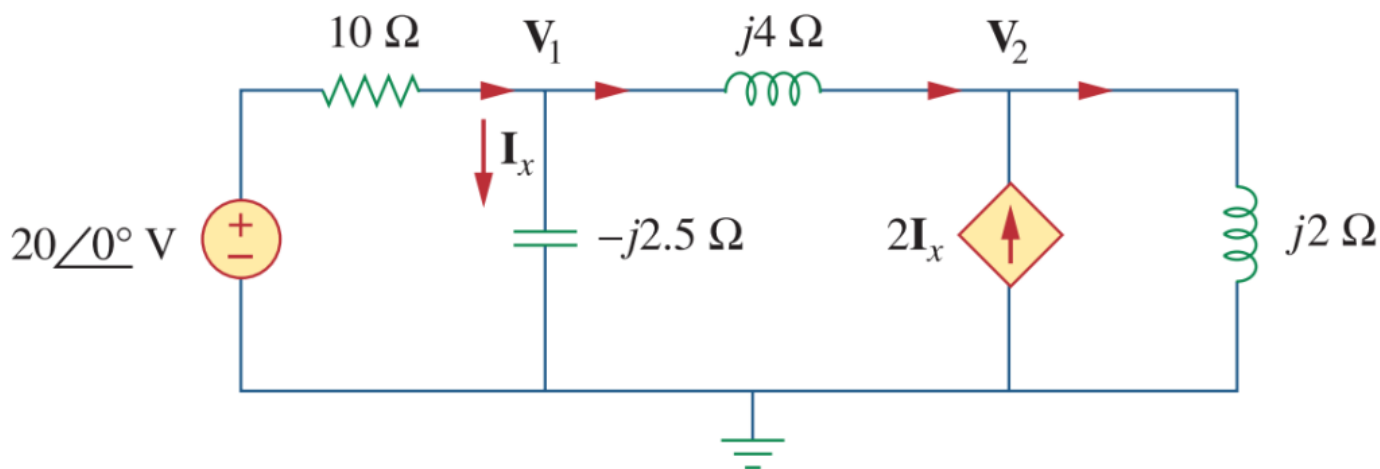
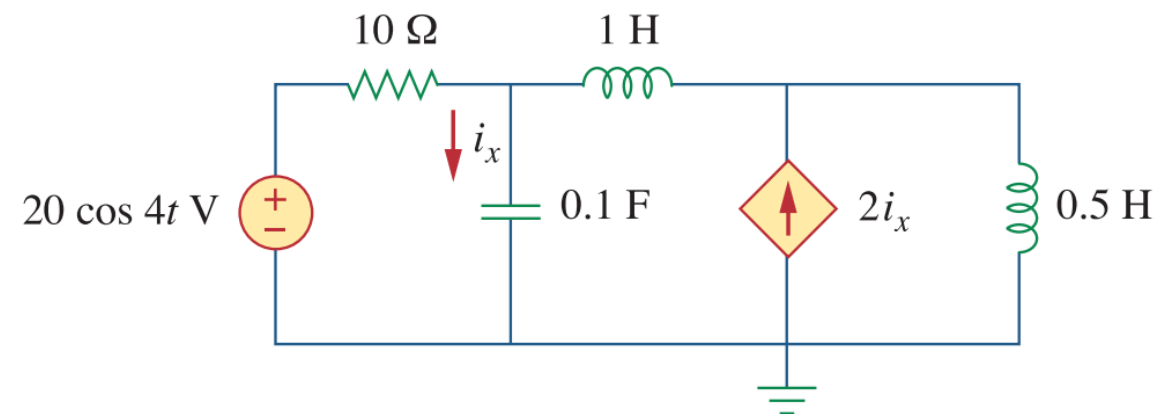
- ❑ Solve the equations obtained in previous steps.
- ❑ Number of equations will be equal to the number of non-reference nodes.

10.1 Nodal and mesh analyses (c)

- Nodal analysis:

Example 10.01: Find $i_x(t)$ in the circuit using nodal analysis.

Solution: (1) Redraw the circuit in phasor domain:



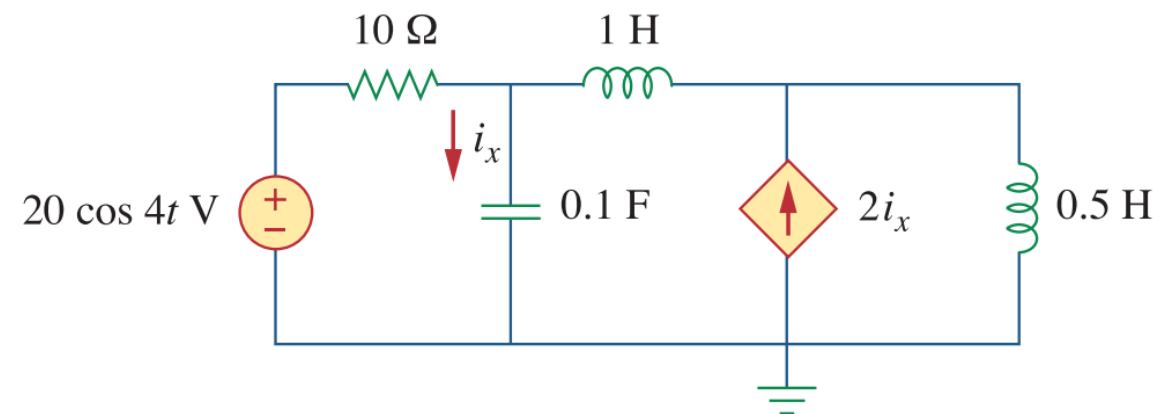
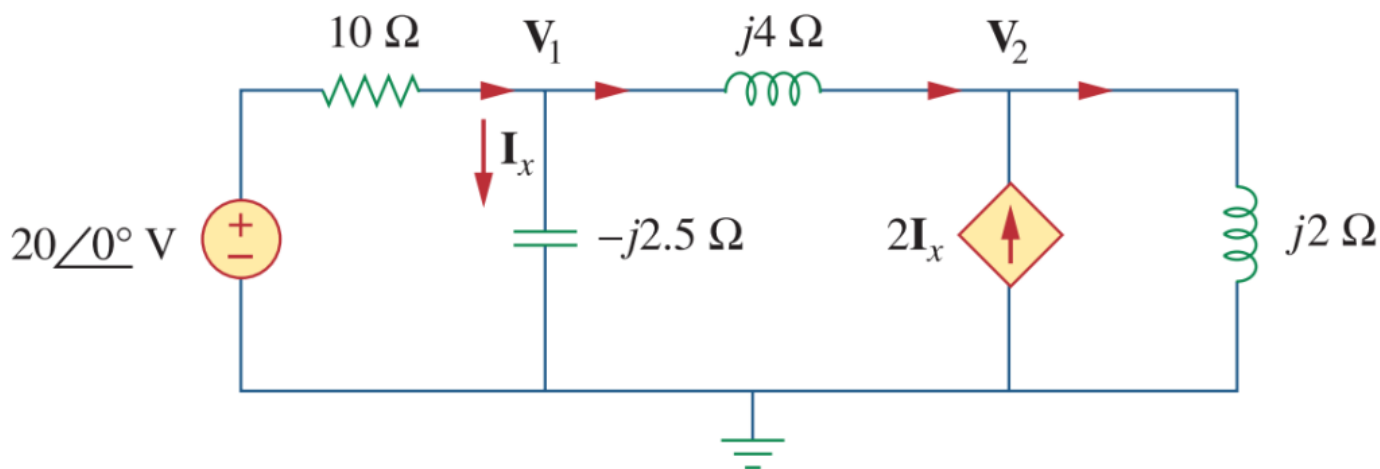
$$\begin{aligned} 20 \cos 4t &\Rightarrow 20\angle 0^\circ, & \omega &= 4 \text{ rad/s} \\ 1 \text{ H} &\Rightarrow j\omega L = j4 \\ 0.5 \text{ H} &\Rightarrow j\omega L = j2 \\ 0.1 \text{ F} &\Rightarrow \frac{1}{j\omega C} = -j2.5 \end{aligned}$$

10.1 Nodal and mesh analyses (d)

- Nodal analysis:

Example 10.01: Find $i_x(t)$ in the circuit using nodal analysis.

Solution: (2) Then we use nodal analysis:



At the V_1 node:

$$\frac{20 - V_1}{10} = \frac{V_1}{-j2.5} + \frac{V_1 - V_2}{j4}$$

At the V_2 node:

$$2I_x + \frac{V_1 - V_2}{j4} = \frac{V_2}{j2}$$

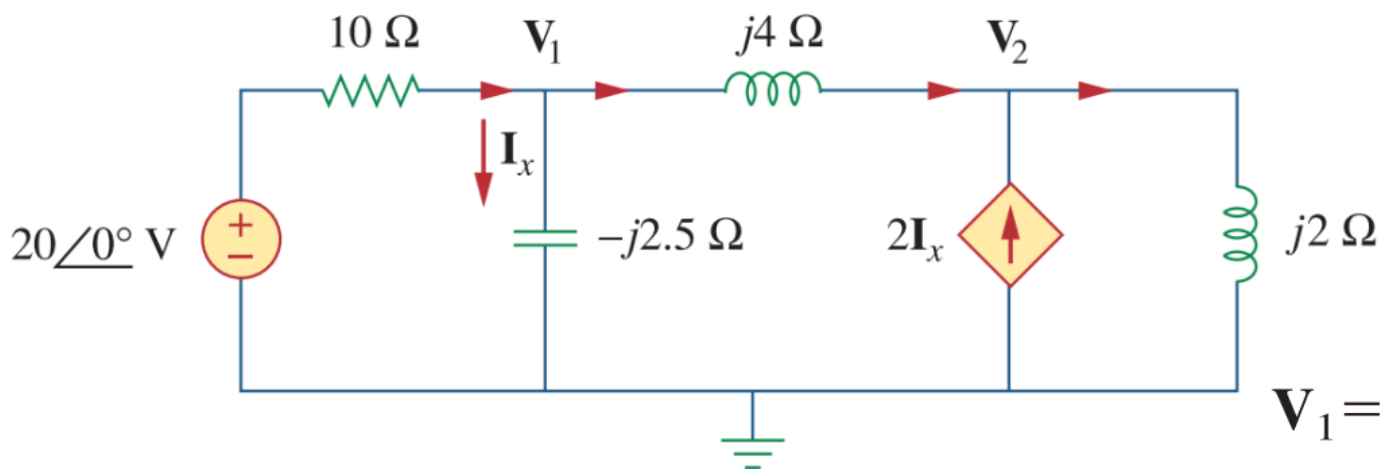
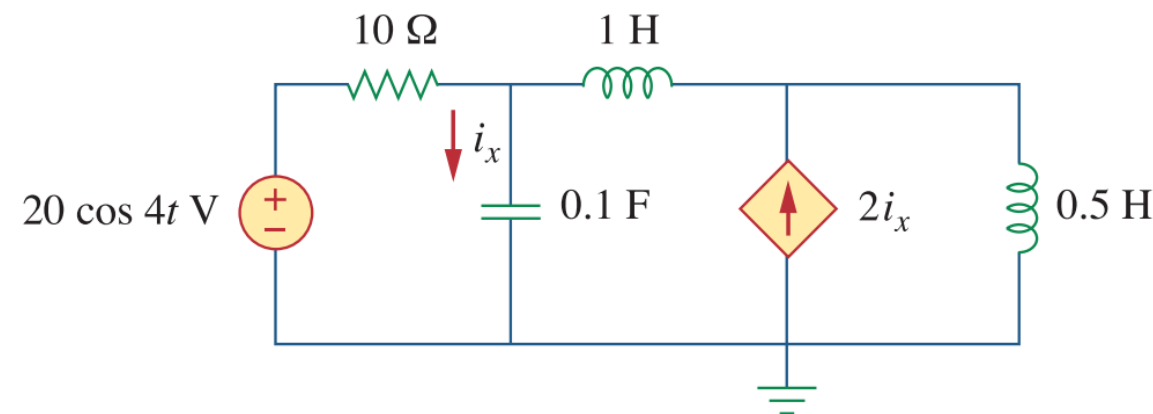
$$I_x = V_1 / -j2.5$$

10.1 Nodal and mesh analyses (e)

- Nodal analysis:

Example 10.01: Find $i_x(t)$ in the circuit using nodal analysis.

Solution: (2) Then we use nodal analysis:



After simplification:

$$\begin{cases} (1 + j1.5)V_1 + j2.5V_2 = 20 \\ 11V_1 + 15V_2 = 0 \end{cases}$$

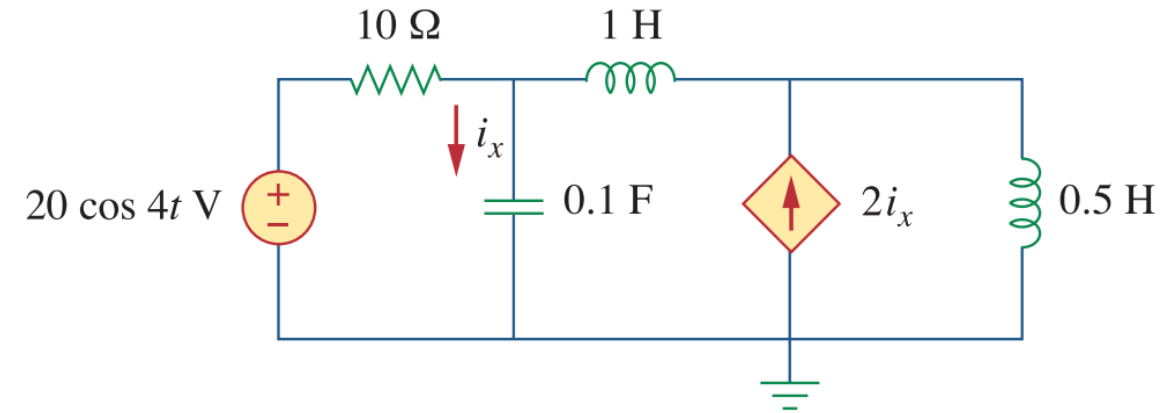
$$\begin{aligned} V_1 &= 18.97 \angle 18.43^\circ \text{ V} \\ V_2 &= 13.91 \angle 198.3^\circ \text{ V} \end{aligned} \quad I_x = \frac{V_1}{-j2.5} = 7.59 \angle 108.4^\circ \text{ A}$$

10.1 Nodal and mesh analyses (f)

- Nodal analysis:

Example 10.01: Find $i_x(t)$ in the circuit using nodal analysis.

Solution: (3) Now back into the time domain:



$$\mathbf{V}_1 = 18.97 \angle 18.43^\circ \text{ V} \quad \Rightarrow \quad v_1(t) = 18.97 \cos(4t + 18.43^\circ) \text{ V}$$

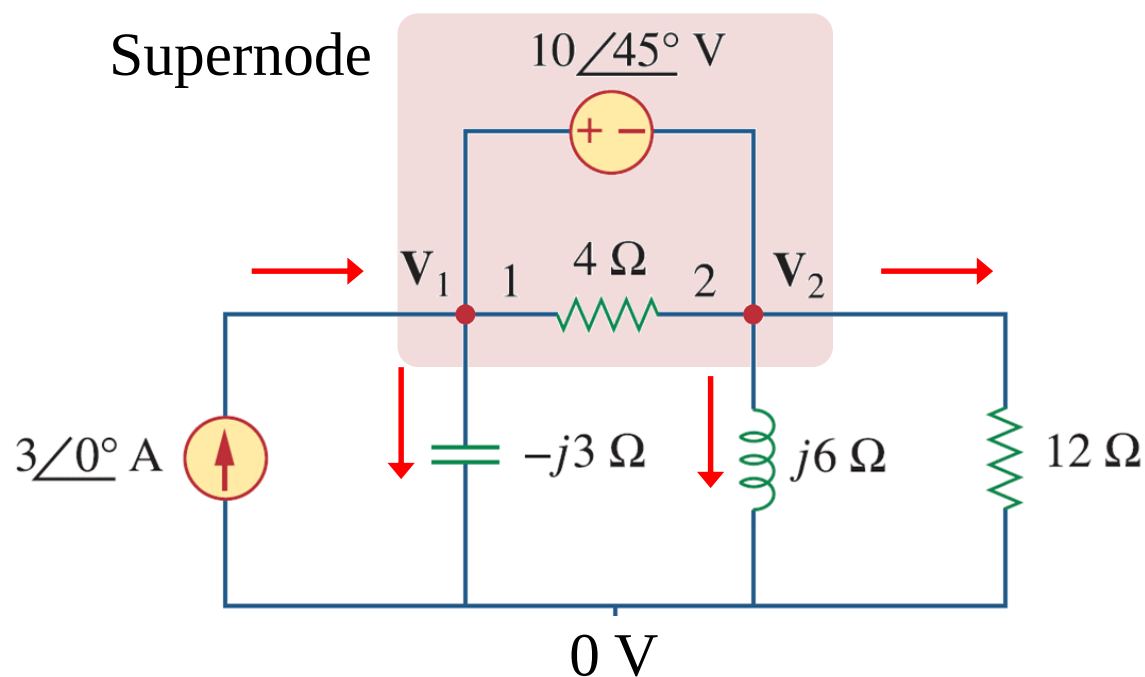
$$\mathbf{V}_2 = 13.91 \angle 198.3^\circ \text{ V} \quad \Rightarrow \quad v_2(t) = 13.91 \cos(4t + 198.3^\circ) \text{ V}$$

$$\mathbf{I}_x = \frac{\mathbf{V}_1}{-j2.5} = 7.59 \angle 108.4^\circ \text{ A} \quad \Rightarrow \quad i_x(t) = 7.59 \cos(4t + 108.4^\circ) \text{ A}$$

10.1 Nodal and mesh analyses (g)

- Nodal analysis:

Example 10.02: compute V_1 and V_2 in the following circuit.



Solution:

- Find the equivalent circuit in phasor domain.
- Use nodal analysis to solve the phasors:

$$\begin{cases} 3 = \frac{V_1}{-j3} + \frac{V_2}{j6} + \frac{V_2}{12} \\ 36 = j4V_1 + (1 - j2)V_2 \\ V_1 = V_2 + 10\angle 45^\circ \end{cases}$$

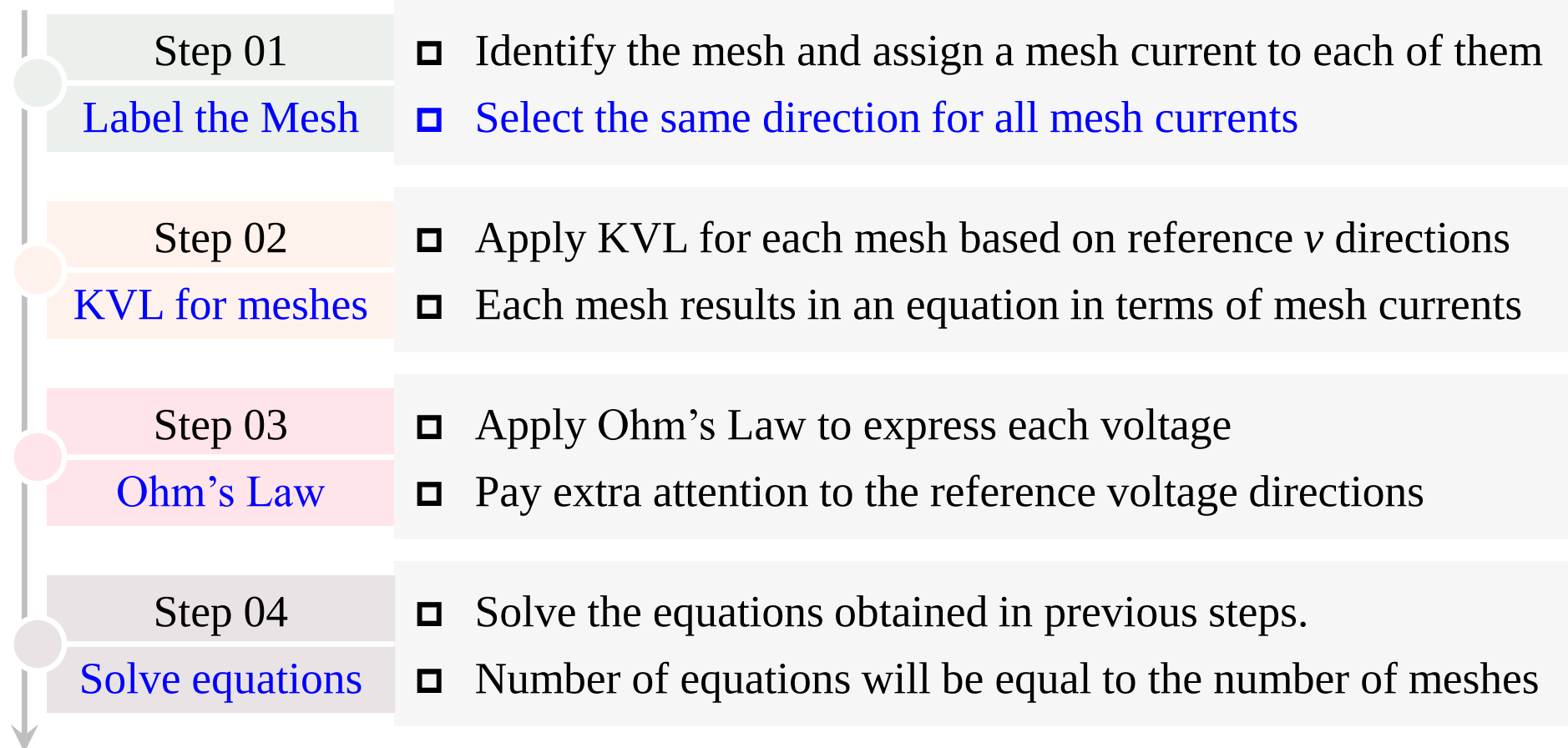
$$\begin{cases} V_2 = 31.41\angle -87.18^\circ \text{ V} \\ V_1 = 25.78\angle -70.48^\circ \text{ V} \end{cases}$$

10.1 Nodal and mesh analyses (h)

- Recall: Mesh analysis

Mesh analysis is a method that is used to solve planar circuits for the currents.

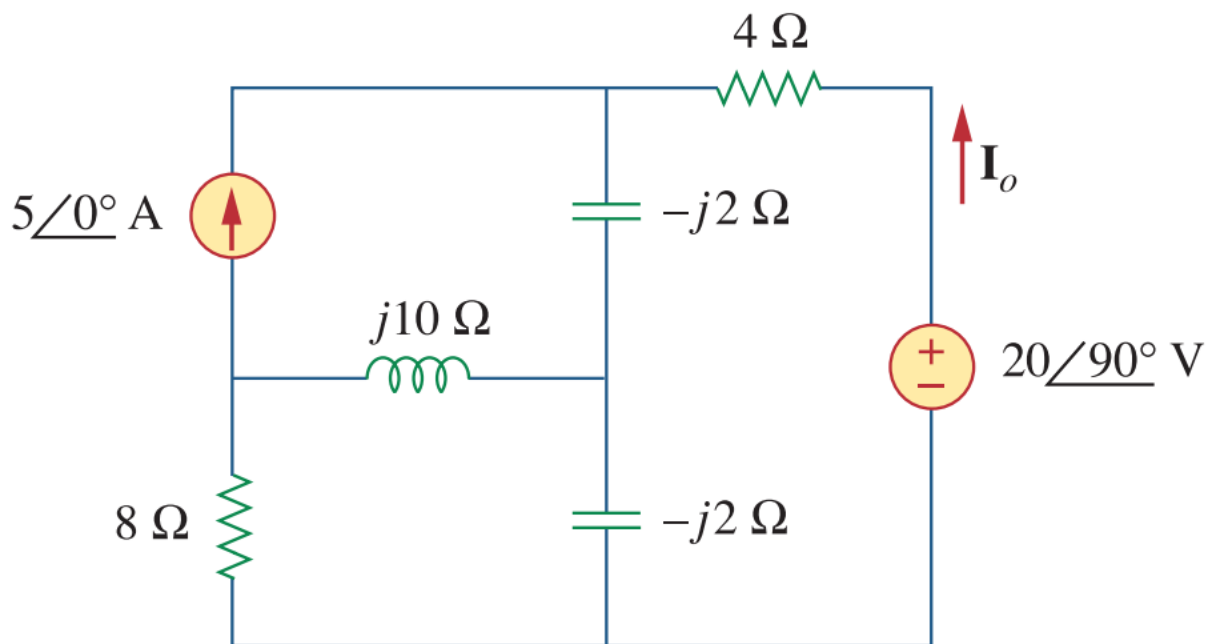
It chooses mesh currents as independent variables, and applied KVL along the meshes.



10.1 Nodal and mesh analyses (i)

- Mesh analysis:

Example 10.03: Determine current $\mathbf{I}_o$ in the circuit using mesh analysis



Solution:

- (1) Redraw the circuit in phasor domain.
- (2) Use mesh analysis:

For mesh 1:

$$(8 + j10 - j2)\mathbf{I}_1 - (-j2)\mathbf{I}_2 - j10\mathbf{I}_3 = 0$$

For mesh 2:

$$(4 - j2 - j2)\mathbf{I}_2 - (-j2)\mathbf{I}_1 - (-j2)\mathbf{I}_3 + 20\angle 90^\circ = 0$$

For mesh 3: $\mathbf{I}_3 = 5$.

$$\begin{cases} \mathbf{I}_2 = 6.12\angle -35.22^\circ \text{ A} \\ \mathbf{I}_o = -\mathbf{I}_2 = 6.12\angle 144.78^\circ \text{ A} \end{cases}$$

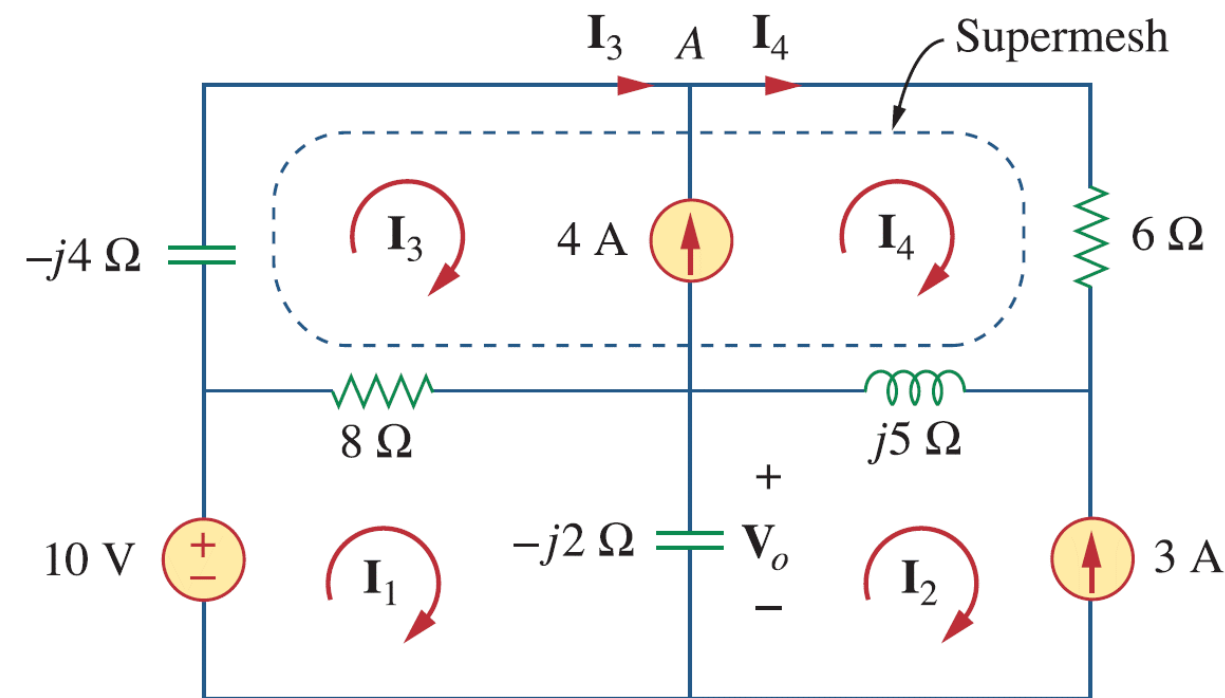
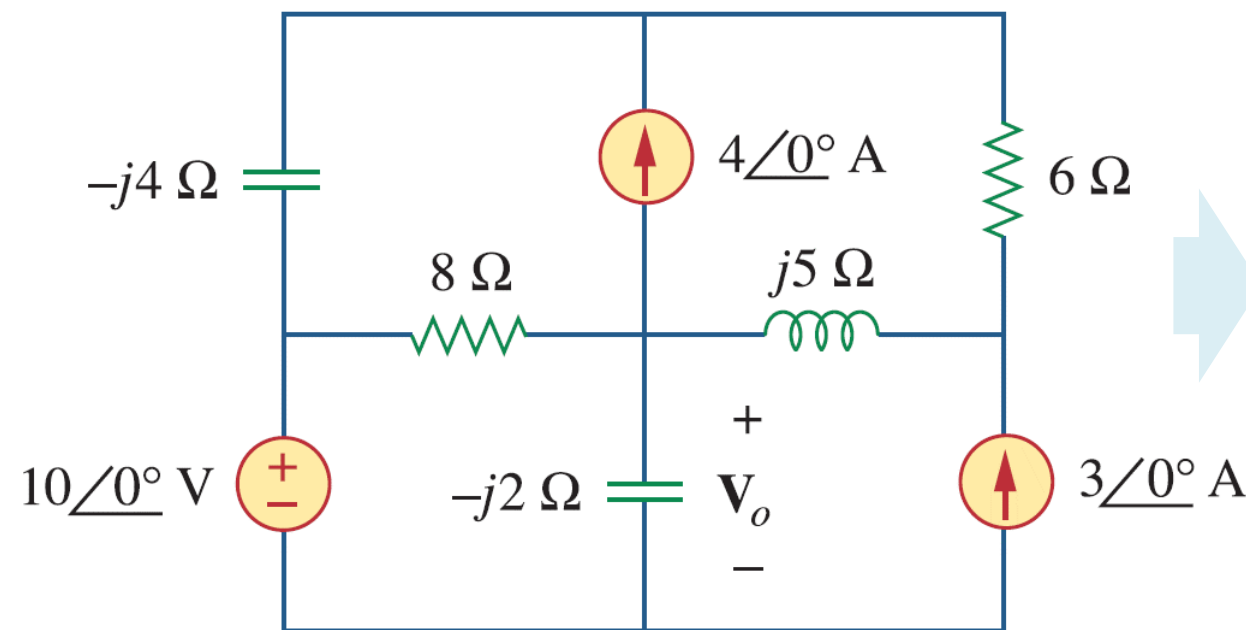
10.1 Nodal and mesh analyses (j)

- Mesh analysis:

Example 10.04: Solve for V_o in the circuit using mesh analysis

Solution:

- (1) Redraw the circuit in phasor domain.
- (2) Use mesh analysis:

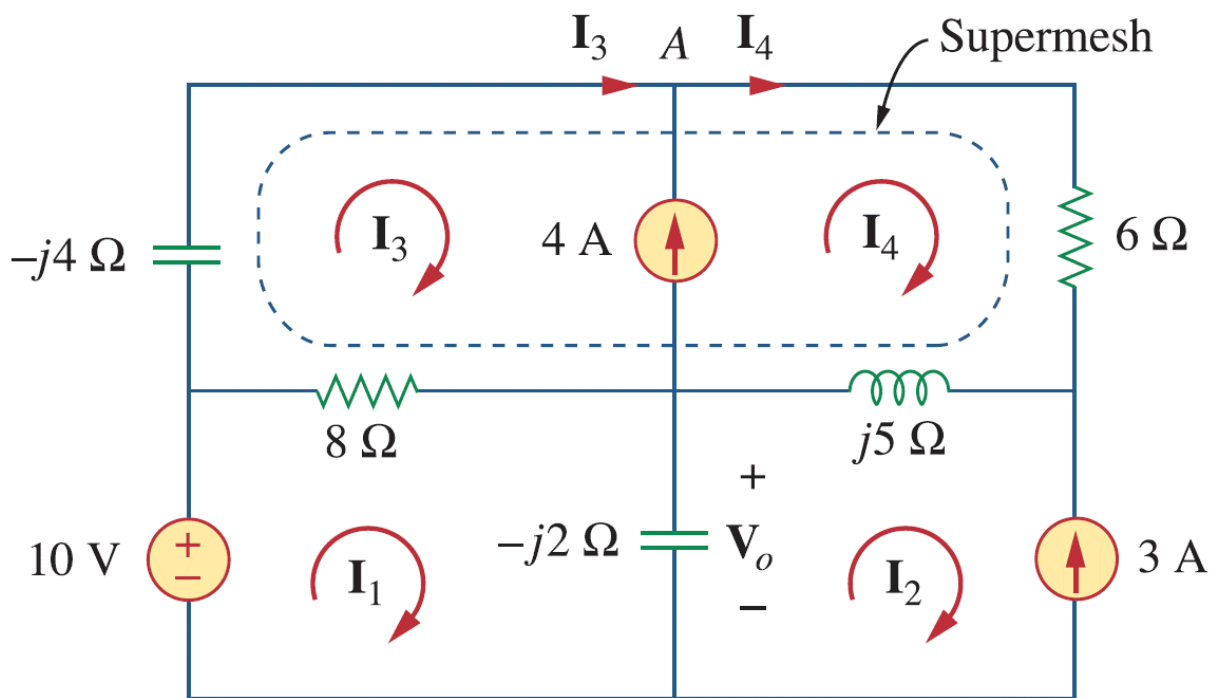


10.1 Nodal and mesh analyses (k)

- Mesh analysis:

Solution (Example 10.04):

- (1) Redraw the circuit in phasor domain.
- (2) Use mesh analysis:



For the mesh 1:

$$-10 + (8 - j2)\mathbf{I}_1 - (-j2)\mathbf{I}_2 - 8\mathbf{I}_3 = 0$$

For the mesh 2:

$$\mathbf{I}_2 = -3$$

For the supermesh:

$$(8 - j4)\mathbf{I}_3 - 8\mathbf{I}_1 + (6 + j5)\mathbf{I}_4 - j5\mathbf{I}_2 = 0$$

For the shared current source:

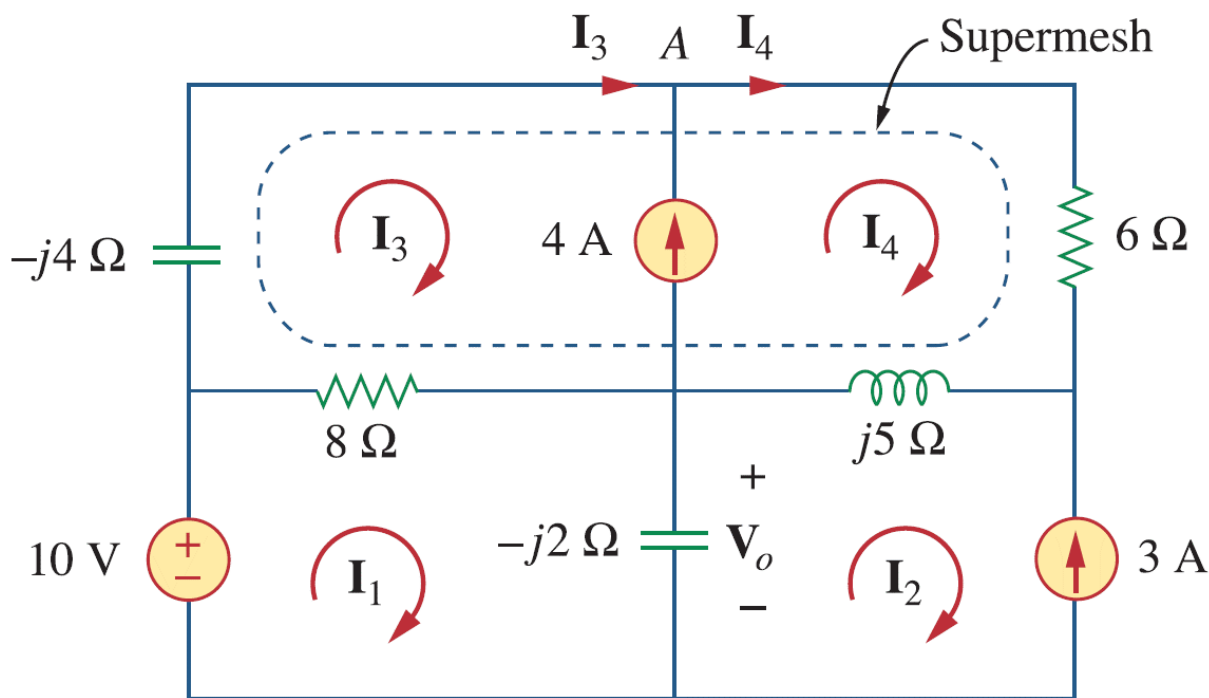
$$\mathbf{I}_4 = \mathbf{I}_3 + 4$$

10.1 Nodal and mesh analyses (I)

- Mesh analysis:

Solution (Example 10.04):

- (1) Find the equivalent circuit in phasor domain.
- (2) Use mesh analysis to solve the phasors:



$$\begin{cases} (8 - j2)\mathbf{I}_1 - 8\mathbf{I}_3 = 10 + j6 \\ -8\mathbf{I}_1 + (14 + j)\mathbf{I}_3 = -24 - j35 \end{cases}$$

$$\begin{aligned} \Rightarrow \mathbf{I}_1 &= 3.618 \angle 274.5^\circ \text{ A} \\ \mathbf{V}_o &= -j2(\mathbf{I}_1 - \mathbf{I}_2) \\ &= -j2(3.618 \angle 274.5^\circ + 3) \\ &= -7.2134 - j6.568 \\ &= 9.756 \angle 222.32^\circ \text{ V} \end{aligned}$$

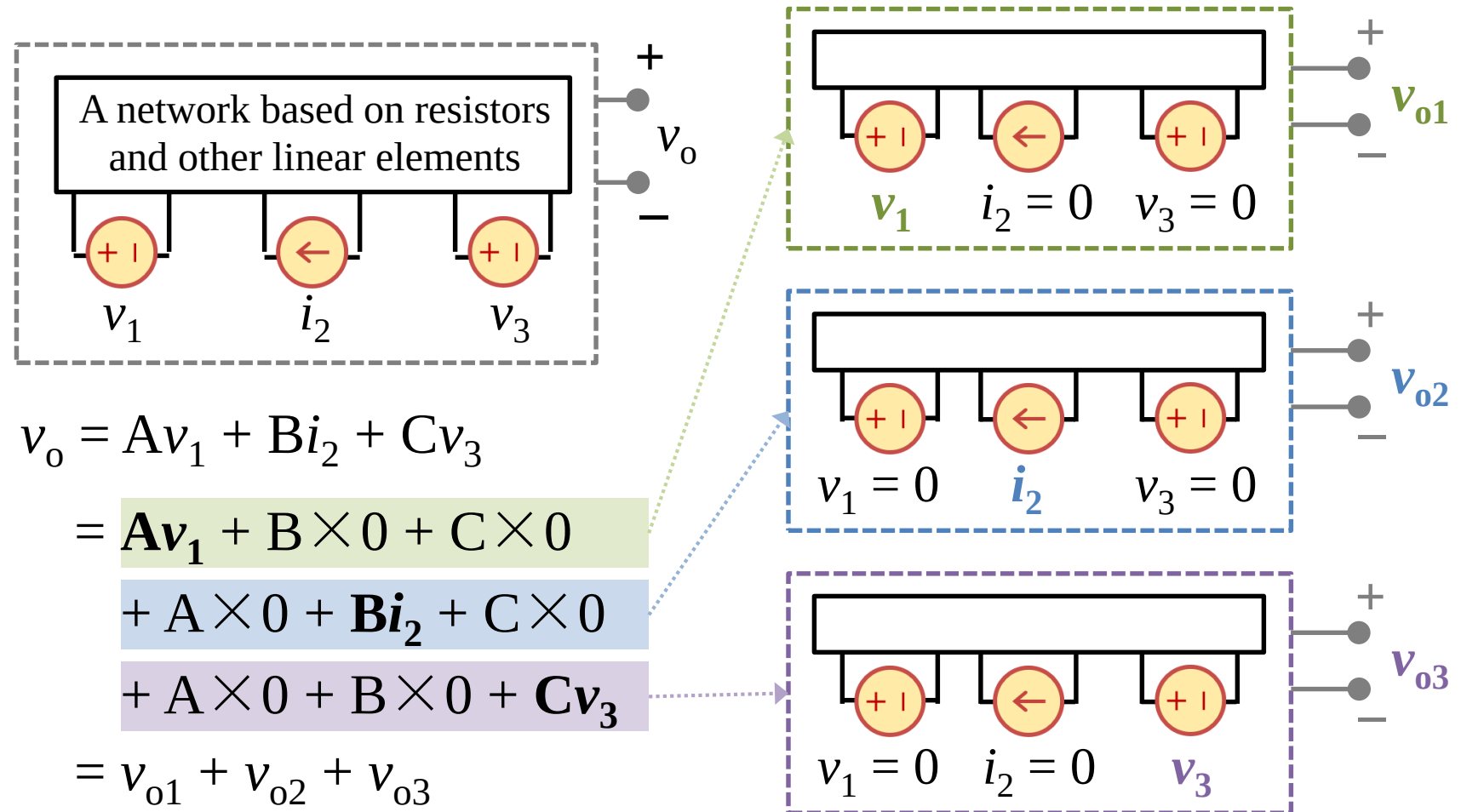
Lesson 10 Steady-state and power analysis of AC circuits

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10.2 Superposition theorem (a)

- Recall: superposition theorem

In a linear system, the **superposition principle** (叠加原理) states that the voltage across (or current through) an element is the algebraic sum of those caused by each independent source acting alone (每个独立源单独作用).



10.2 Superposition theorem (b)

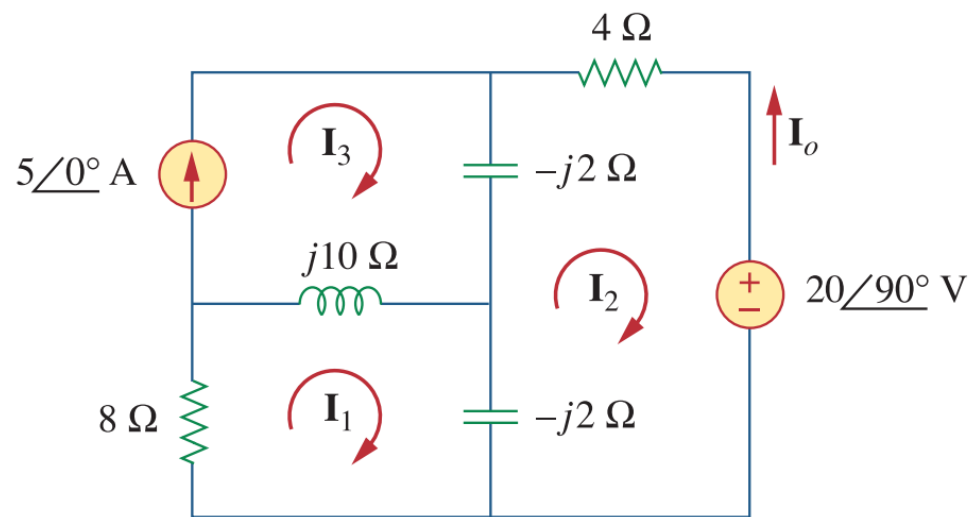
- Superposition theorem for AC circuits

- a) Since AC circuits based on linear elements, independent sources, and linear dependent sources, it is also possible to apply the principle of superposition.
- b) Particular attention is needed when the circuit has various sources that operate at different frequencies. In such case, corresponding to each source, the circuit would have distinct equivalents in the frequency (phasor) domain.
- c) This is because:
 - The circuit elements may have a frequency-dependent impedances.
 - The resulting voltages/ currents corresponding to each frequencies must be converted back to the time domain before they can be added together. This is because there is an exponential factor $e^{j\omega t}$ implicit in the sinusoidal analysis.

10.2 Superposition theorem (c)

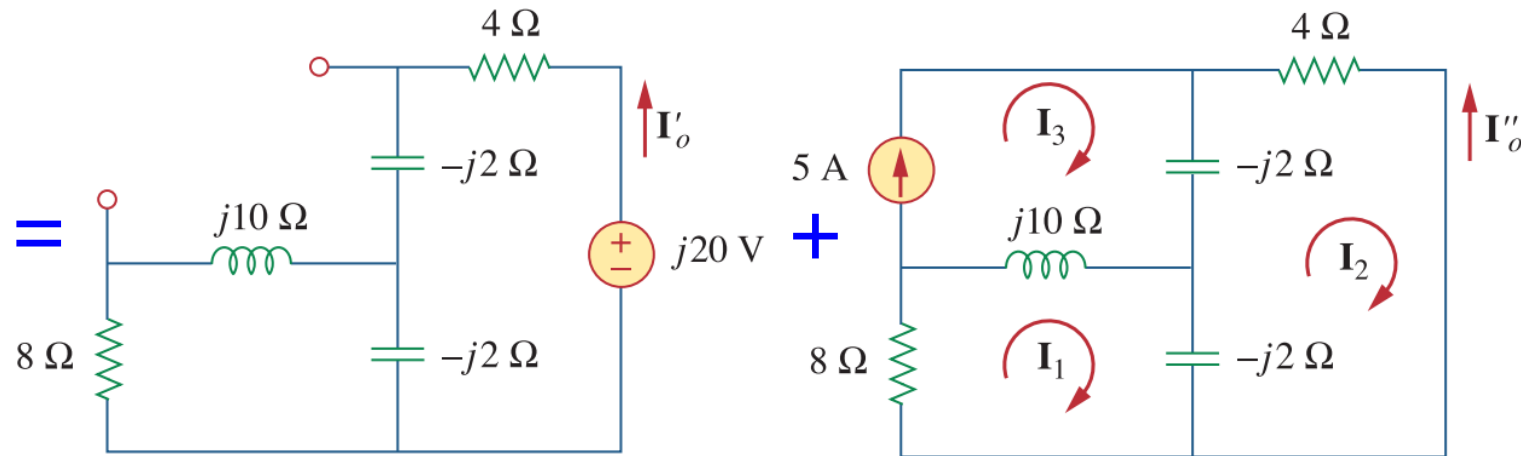
- Example 10.05

Example 10.05: use superposition theorem to find $\mathbf{I}_o$ in the following circuit.



Solution:

- (1) Redraw the circuit in phasor domain.
- (2) Use superposition:



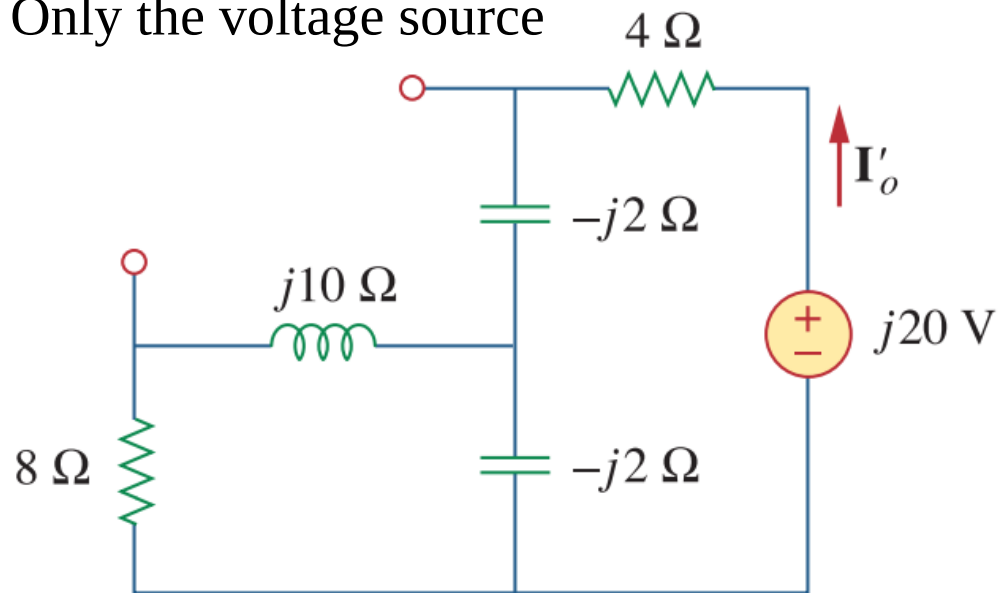
$$\mathbf{I}_o = \mathbf{I}'_o + \mathbf{I}''_o$$

10.2 Superposition theorem (d)

• Example 10.05

Example 10.05: use superposition theorem to find $\mathbf{I}_o$ in the following circuit.

Only the voltage source



Solution:

- (1) Find the equivalent circuit in phasor domain.
- (2) Use superposition to solve the phasors:

$$\begin{aligned}\mathbf{Z} &= 4 - j2 + (8 + j10) \parallel (-j2) \\ &= 4 - j2 - \frac{2j(8 + j10)}{8 + j10 - 2j} \\ &= 0.25 - j2.25 \Omega\end{aligned}$$

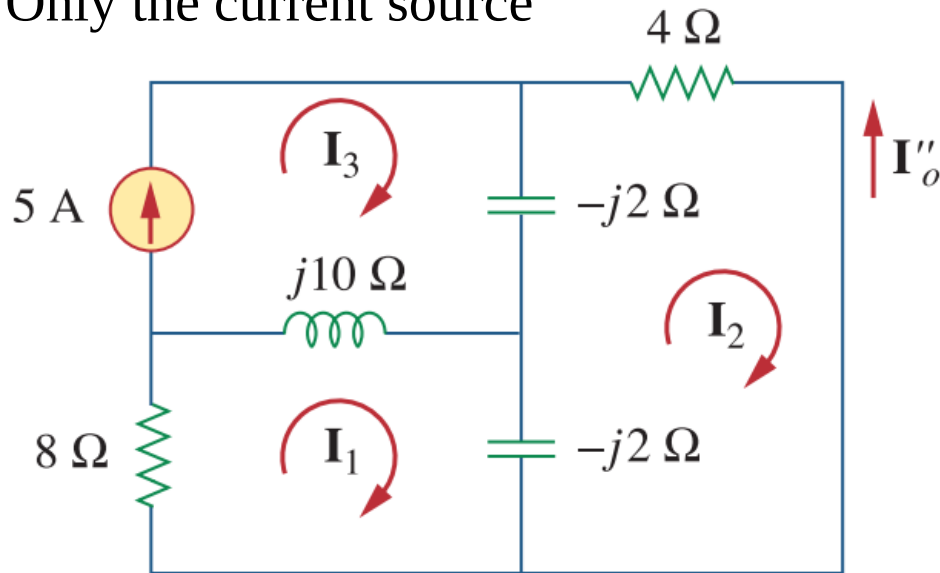
$$\mathbf{I}'_o = \frac{j20}{0.25 - j2.25} = -2.353 + j2.353$$

10.2 Superposition theorem (e)

• Example 10.05

Example 10.05: use superposition theorem to find $\mathbf{I}_o$ in the following circuit.

Only the current source



Solution:

- (1) Find the equivalent circuit in phasor domain.
- (2) Use superposition to solve the phasors:

$$\begin{cases} \text{For mesh 1: } (8 + j8)\mathbf{I}_1 - j10\mathbf{I}_3 + j2\mathbf{I}_2 = 0 \\ \text{For mesh 2: } (4 - j4)\mathbf{I}_2 + j2\mathbf{I}_1 + j2\mathbf{I}_3 = 0 \\ \text{For mesh 3: } \mathbf{I}_3 = 5 \end{cases}$$

$$\mathbf{I}_o'' = -\mathbf{I}_2 = -2.647 + j1.176$$

$$\mathbf{I}_o = \mathbf{I}_o' + \mathbf{I}_o'' = -5 + j3.529 = 6.12 \angle 144.78^\circ \text{ A}$$

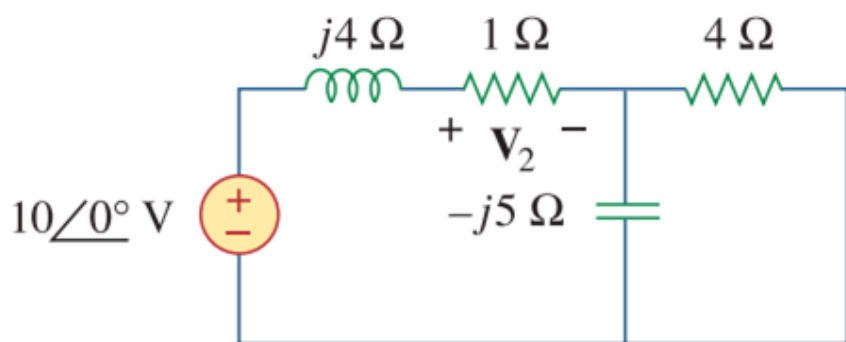
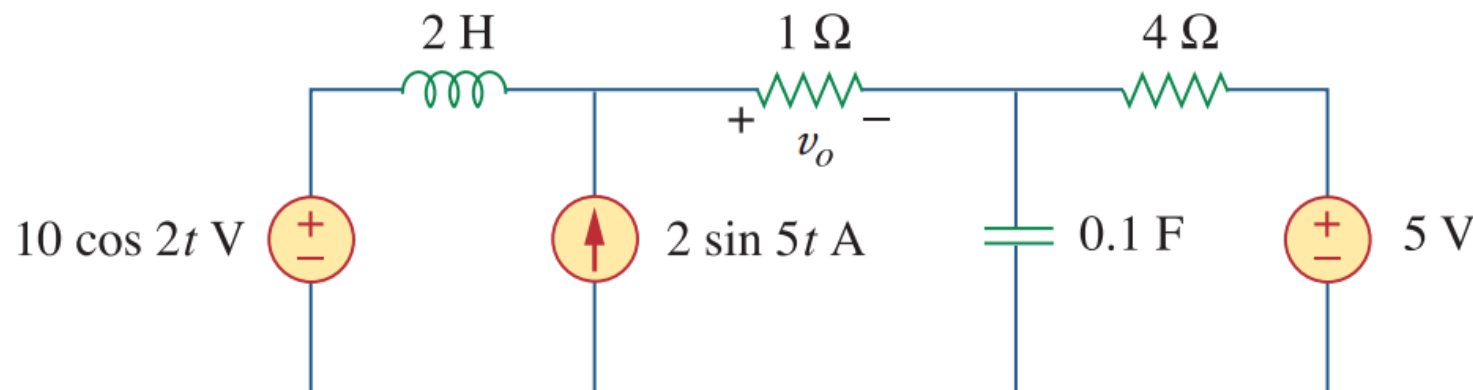
10.2 Superposition theorem (f)

• Example 10.06

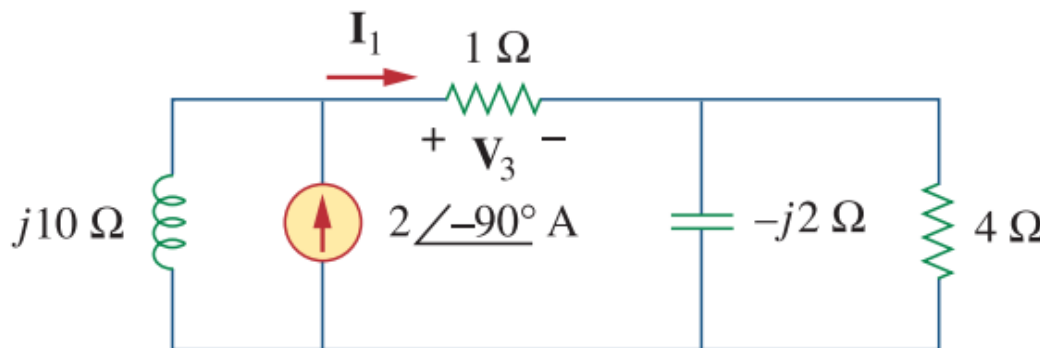
Example 10.06: Determine the $v_o(t)$.

Solution:

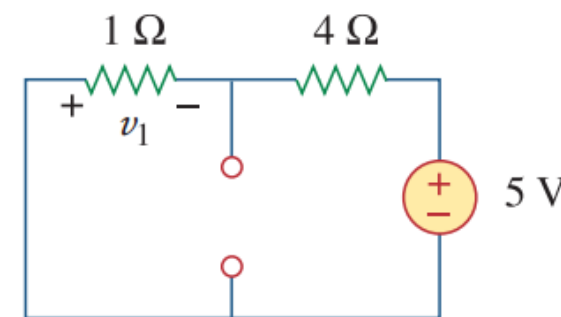
- (1) Find the equivalent circuit in phasor domain & use superposition to solve the phasors:
- (2) Impedances can be different with different frequencies.



$\omega = 2$



$\omega = 5$

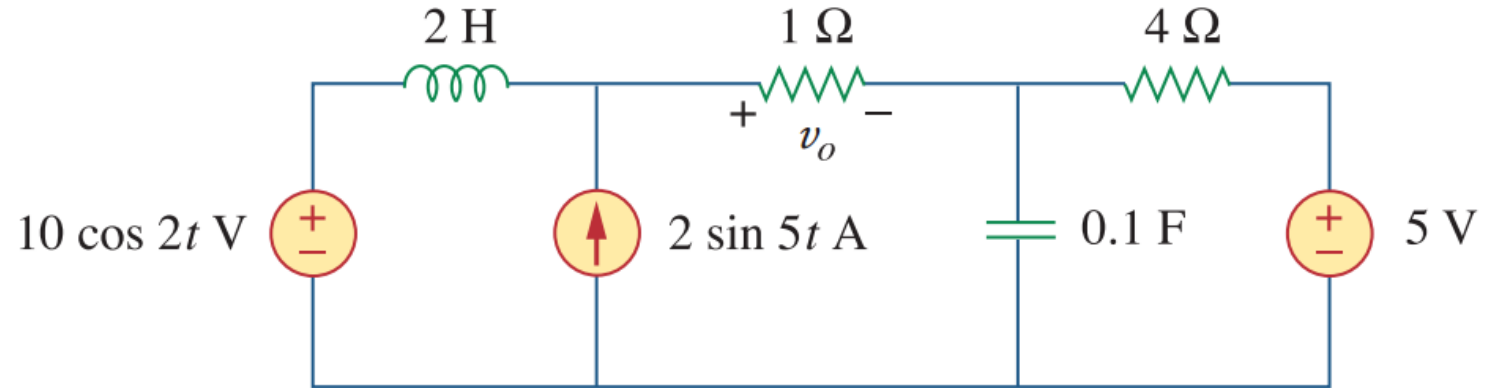


DC

10.2 Superposition theorem (g)

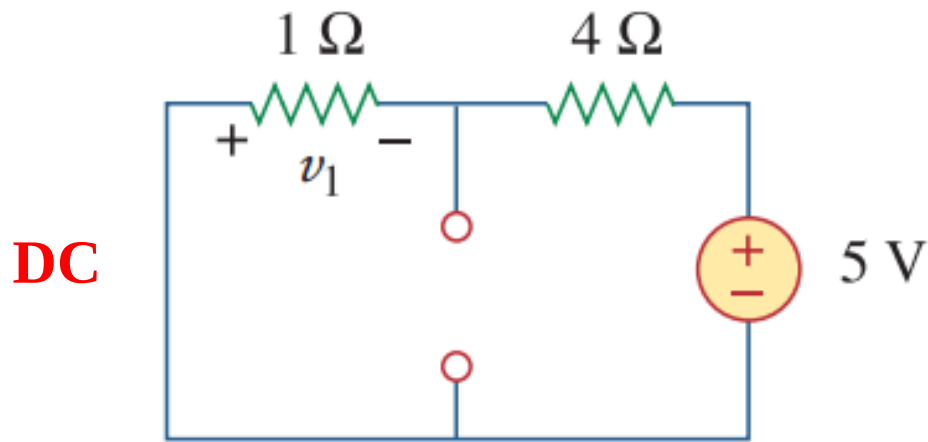
- Example 10.06

Example 10.06: Determine the $v_o(t)$.



Solution:

(3) When only the DC voltage source is ON:



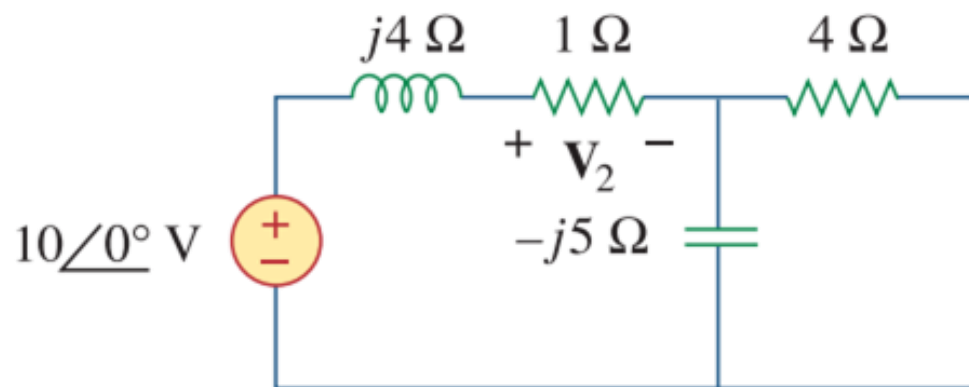
$$-v_1 = \frac{1}{1 + 4}(5) = 1 \text{ V}$$

$$v_1 = -1 \text{ V}$$

10.2 Superposition theorem (h)

• Example 10.06

Solution: (4) When only the AC voltage source is ON



$$\begin{aligned} 10 \cos 2t &\Rightarrow 10\angle 0^\circ, & \omega &= 2 \text{ rad/s} \\ 2 \text{ H} &\Rightarrow j\omega L = j4 \Omega \\ 0.1 \text{ F} &\Rightarrow \frac{1}{j\omega C} = -j5 \Omega \end{aligned}$$

$$\begin{aligned} \mathbf{Z} &= j4 + 1 + 4 \parallel (-j5) \\ &= 4 + 1 - \frac{20j}{4 - 5j} \\ &= 3.439 - j2.049 \Omega \end{aligned}$$

Based on voltage division:

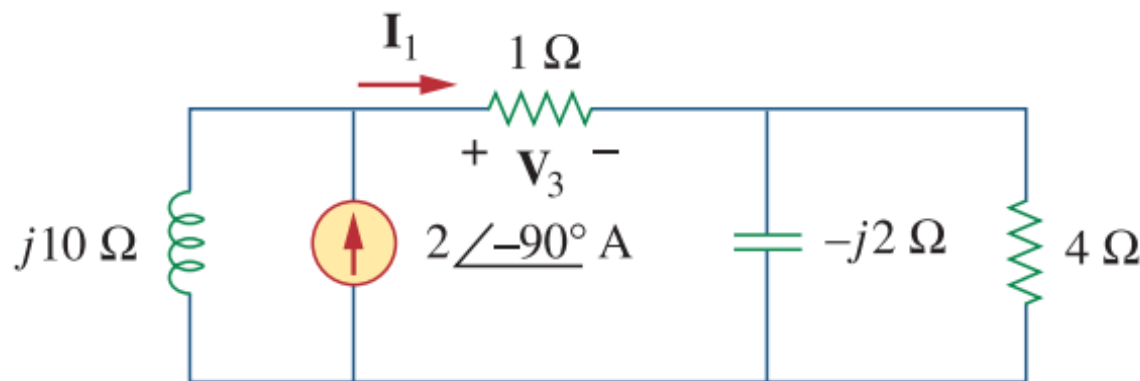
$$\mathbf{V}_2 = \frac{1}{\mathbf{Z}} 10\angle 0^\circ = \frac{10}{3.439 - j2.049} = 2.498\angle -30.79^\circ$$

$$v_2 = 2.498 \cos(2t - 30.79^\circ)$$

10.2 Superposition theorem (i)

• Example 10.06

Solution: (5) When only the AC current source is ON



$$\begin{aligned}
 2 \sin 5t &\Rightarrow 2 \angle -90^\circ, & \omega &= 5 \text{ rad/s} \\
 2 \text{ H} &\Rightarrow j\omega L = j10 \Omega \\
 0.1 \text{ F} &\Rightarrow \frac{1}{j\omega C} = -j2 \Omega
 \end{aligned}$$

$$\begin{aligned}
 \mathbf{Z}_1 &= 1 + 4 \parallel (-j8) \\
 &= 1 - \frac{j8}{4 - j2} \\
 &= 0.8 - j1.6 \Omega
 \end{aligned}$$

Based on current division:

$$\mathbf{I}_1 = \frac{j10}{\mathbf{Z}_1 + j10} 2 \angle -90^\circ = \frac{j10}{\mathbf{Z}_1 + j10} (-2j)$$

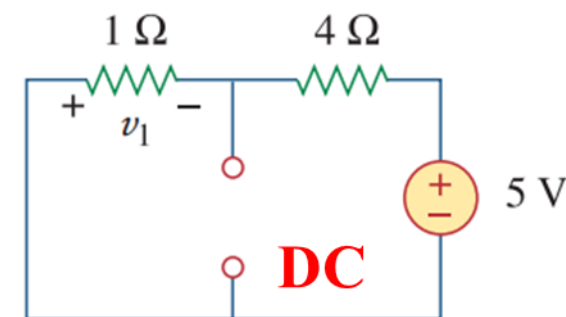
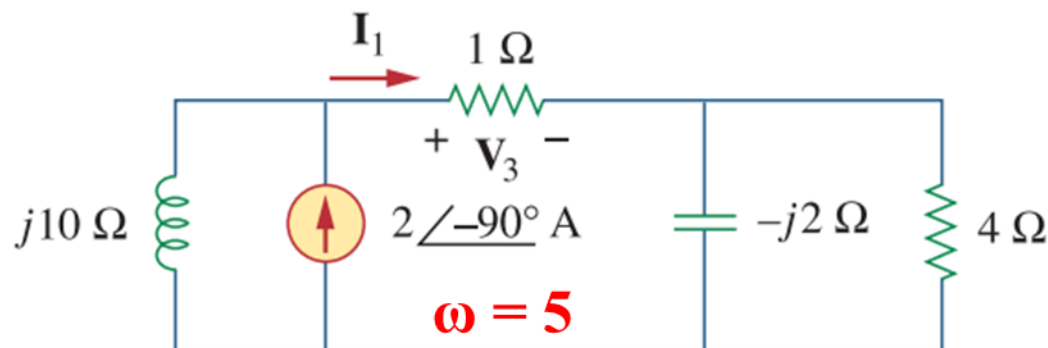
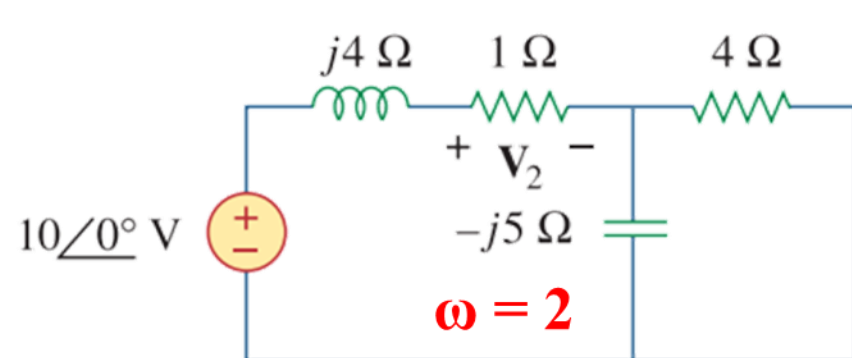
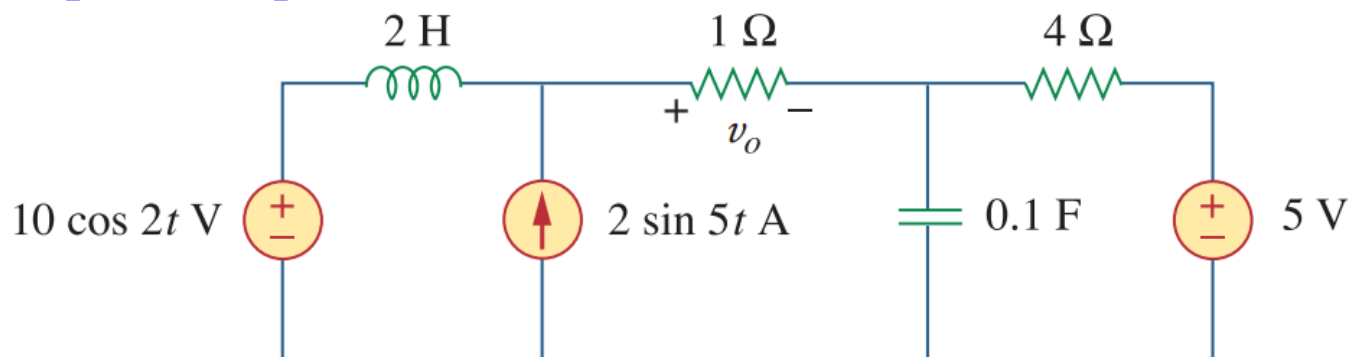
$$\mathbf{V}_3 = \mathbf{I}_1 \times 1 = \frac{j10}{1.8 + j8.4} (-j2) = 2.328 \angle -80^\circ \text{ V}$$

$$v_3 = 2.33 \cos(5t - 80^\circ) = 2.33 \sin(5t + 10^\circ) \text{ V}$$

10.2 Superposition theorem (j)

- Example 10.06

Solution: (6) Sum up the response in time domain



$$v_o(t) = -1 + 2.498 \cos(2t - 30.79^\circ) + 2.33 \sin(5t + 10^\circ) \text{ V}$$

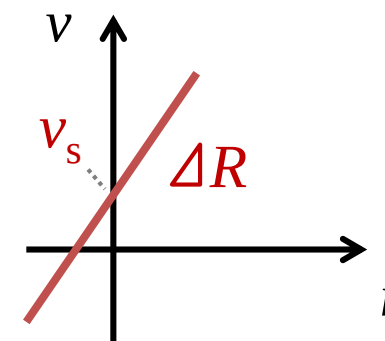
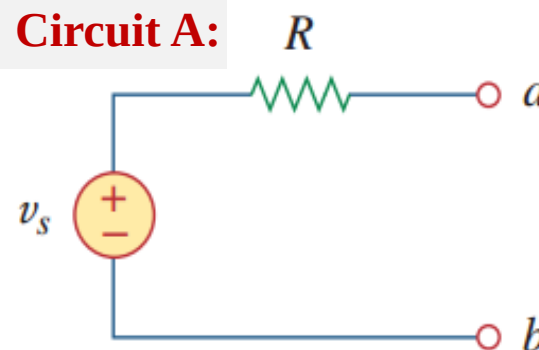
10.3 Source transformation (a)

- Recall: Source transformation

Source transformation (电源变换)

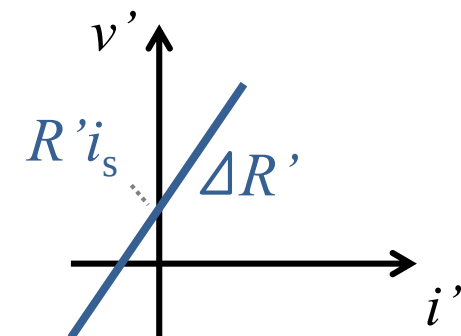
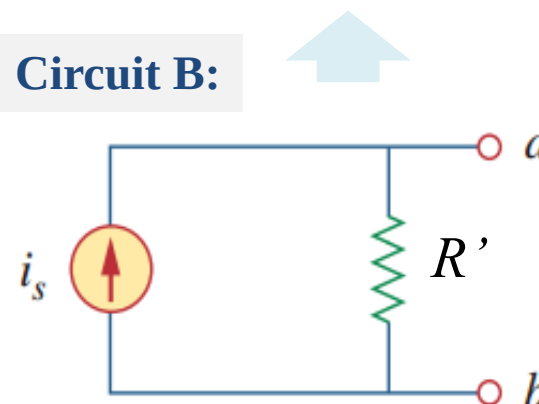
refers to the process of replacing a voltage source in series with a resistor by an equivalent circuit including a current source and a resistor, or vice versa.

Circuit A:



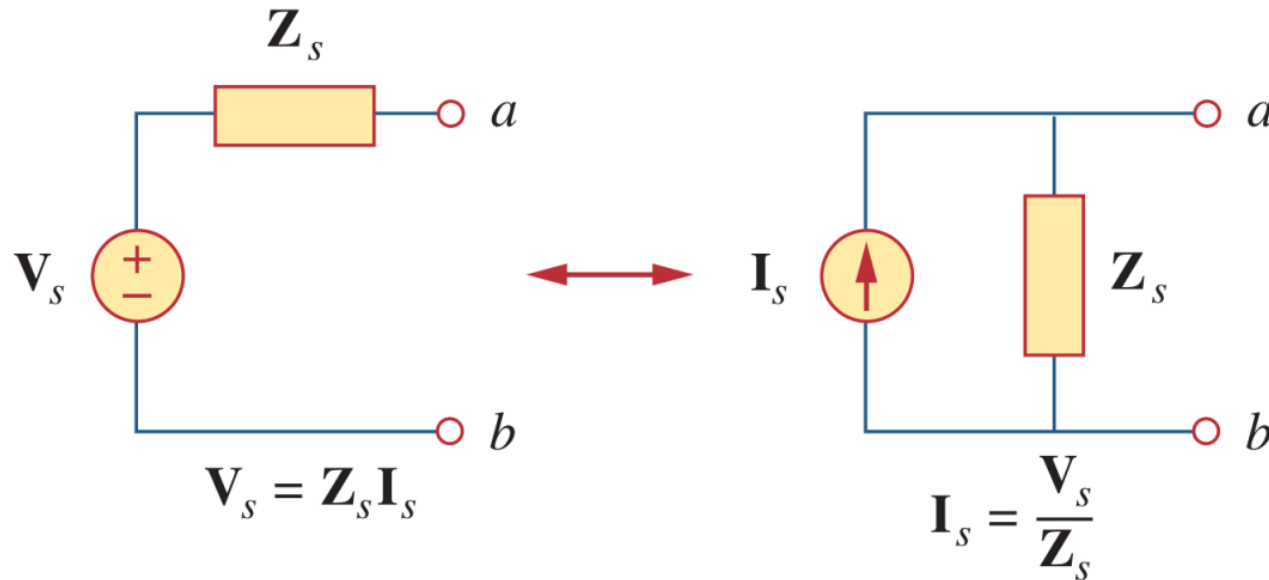
To make them equivalent: (1) $R = R'$ & (2) $v_s = R \cdot i_s$

Circuit B:



10.3 Source transformation (b)

- Source transformation for AC circuits



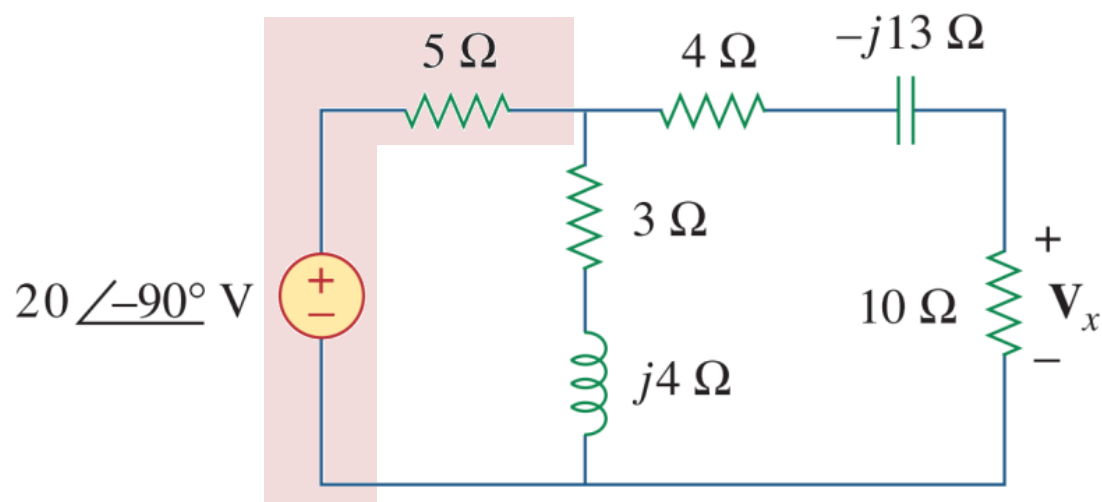
$$V_s = Z_s I_s \quad \Leftrightarrow \quad I_s = \frac{V_s}{Z_s}$$

- a) Same as the case with DC sources, source transformation of circuits excited by AC sources in frequency domain involves transforming a voltage source in series with an impedance to a current source in parallel with an impedance.

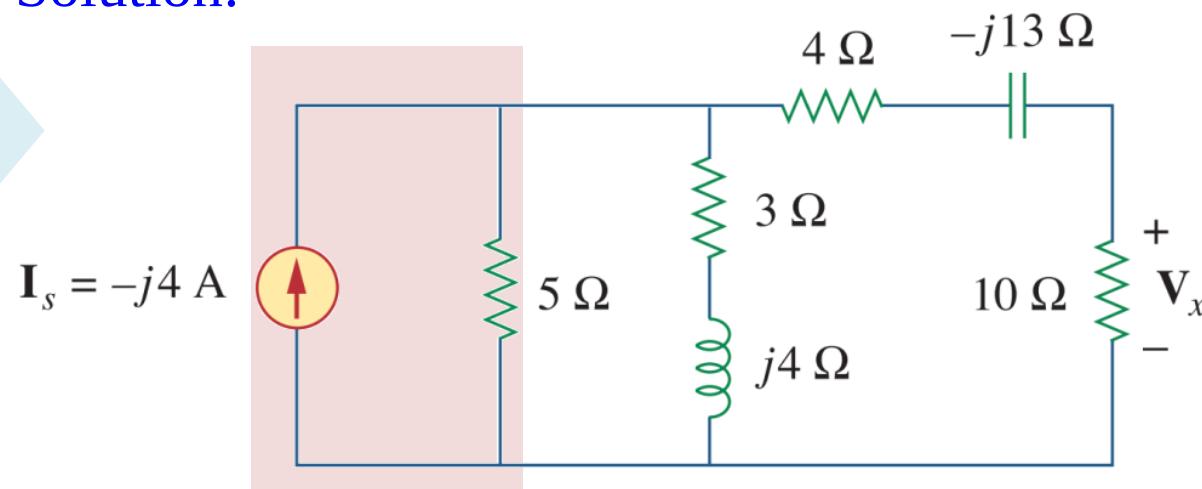
10.3 Source transformation (c)

- Example 10.07

Example 10.07: Calculate $\mathbf{V}_x$ in the circuit using the concept of source transformation.



Solution:



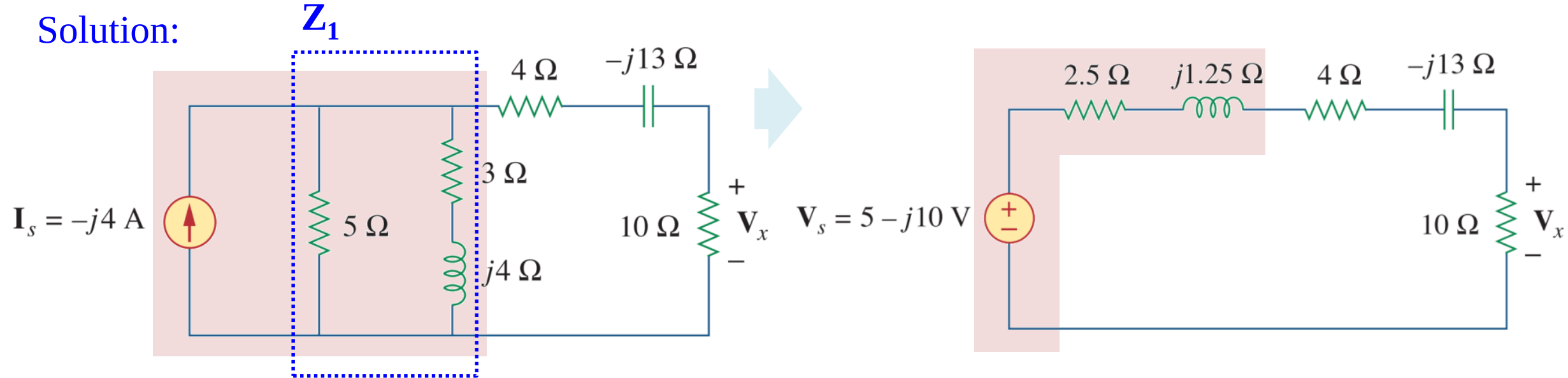
$$\mathbf{I}_s = \frac{20 \angle -90^\circ}{5} = 4 \angle -90^\circ = -j4 \text{ A}$$

10.3 Source transformation (d)

- Example 10.07

Example 10.07: Calculate $\mathbf{V}_x$ in the circuit using the concept of source transformation.

Solution:



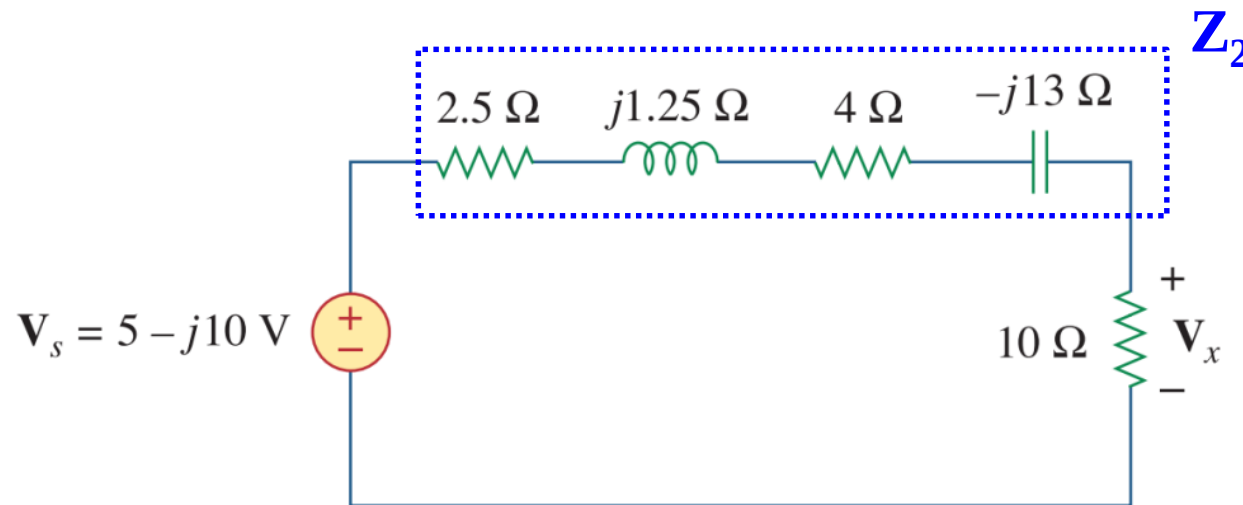
$$\mathbf{Z}_1 = \frac{5(3 + j4)}{8 + j4} = 2.5 + j1.25 \, \Omega \quad \mathbf{V}_s = \mathbf{I}_s \mathbf{Z}_1 = -j4(2.5 + j1.25) = 5 - j10 \, \text{V}$$

10.3 Source transformation (e)

- Example 10.07

Example 10.07: Calculate V_x in the circuit using the concept of source transformation.

Solution:



$$\begin{aligned} Z_2 &= 2.5 + j1.25 + 4 - j13 \\ &= 6.5 - j11.75 \Omega \end{aligned}$$

Is this impedance capacitive or inductive?

$$\begin{aligned} V_x &= \frac{10}{10 + Z_2} (5 - j10) \\ &= 5.519 \angle -28^\circ \end{aligned}$$

Does V_x lead or lag the V_s ?

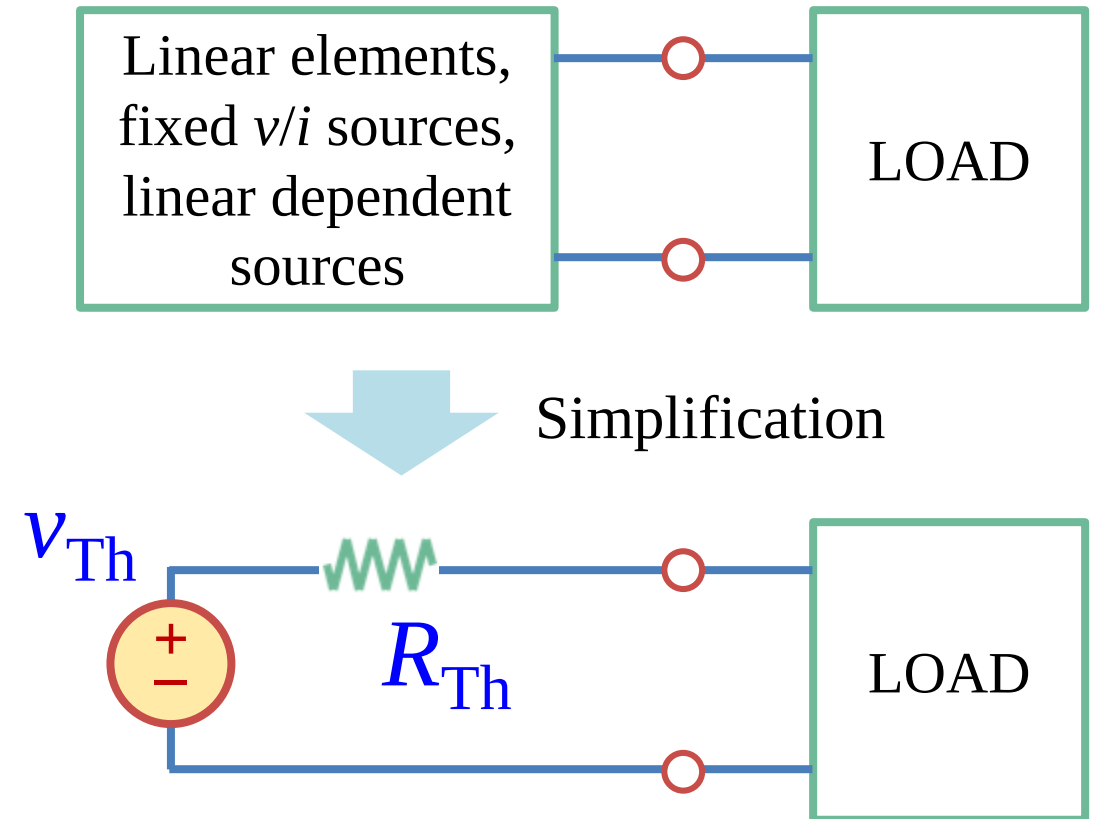
Lesson 10 Steady-state and power analysis of AC circuits

10.1	Nodal and mesh analyses	Slides 13-24
10.2-10.3	Superposition theorem and source transformation	Slides 26-40
10.4	Thevenin and Norton equivalent circuits	Slides 42-52
10.5	AC circuits with op amp	Slides 54-58
10.6	AC power in time domain	Slides 60-71
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10.4 Thevenin's and Norton's theorems (a)

- Recall: Thevenin's theorem

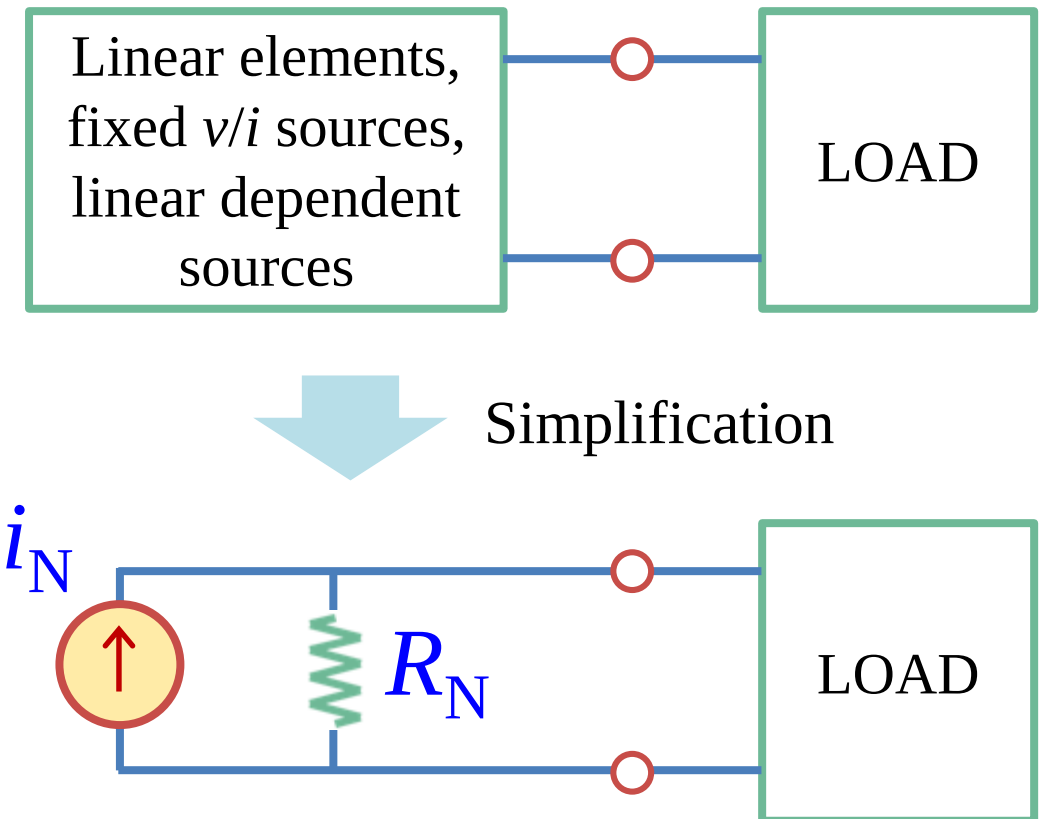
- a) Thévenin's theorem:** any two-terminal network consisting of linear elements, fixed voltage/current sources and linear dependent sources is externally equivalent to a circuit consisting of a resistor (R_{Th}) in series with a fixed voltage source (v_{Th}).
- b) v_{Th}** is the open-circuit voltage at the terminals.
- c) R_{Th}** is the input or the equivalent resistance measured at the terminals when all the internal independent sources are turned off.



10.4 Thevenin's and Norton's theorems (b)

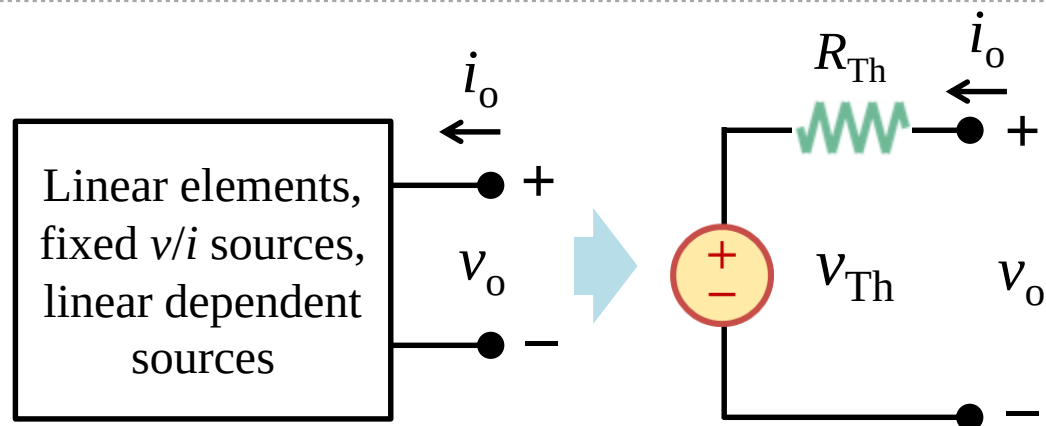
- Recall: Norton's theorem

- a) Norton's theorem:** Any two-terminal network consisting of resistors, fixed voltage/current sources and linear dependent sources is externally equivalent to a circuit consisting of a resistor (R_N) in parallel with a fixed current source (i_N).
- b) i_N** is the short-circuit current at the terminals.
- c) R_N** is the input or the equivalent resistance measured at the terminals when all the internal independent sources are turned off.



10.4 Thevenin's and Norton's theorems (c)

Thévenin's theorem



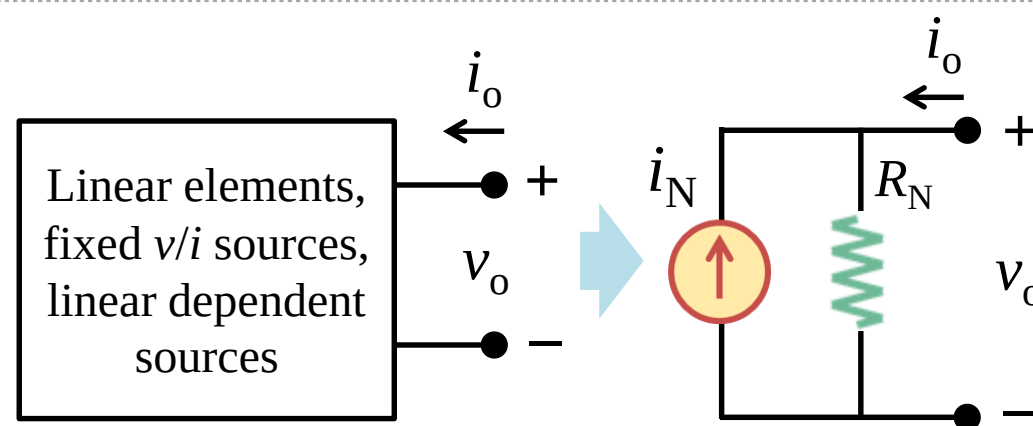
$$v_o = R_{Th} \cdot i_o + V_{Th}$$

Open-circuit voltage: $v_{OC} = v_{Th}$

Short-circuit current: $i_{SC} = v_{Th}/R_{Th} = -i_o$

Equivalent R : $R_{Th} = v_{OC}/i_{SC} = R_{eq}$

Norton's theorem



$$i_o = v_o/R_N - i_N$$

Open-circuit voltage: $v_{OC} = i_N \cdot R_N$

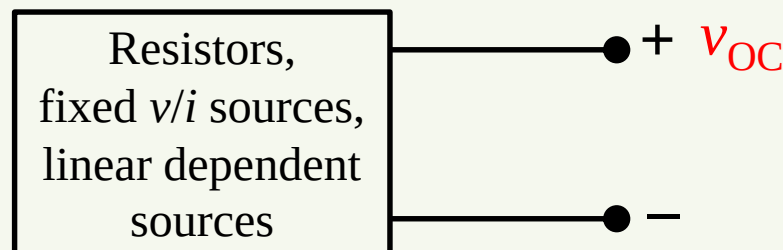
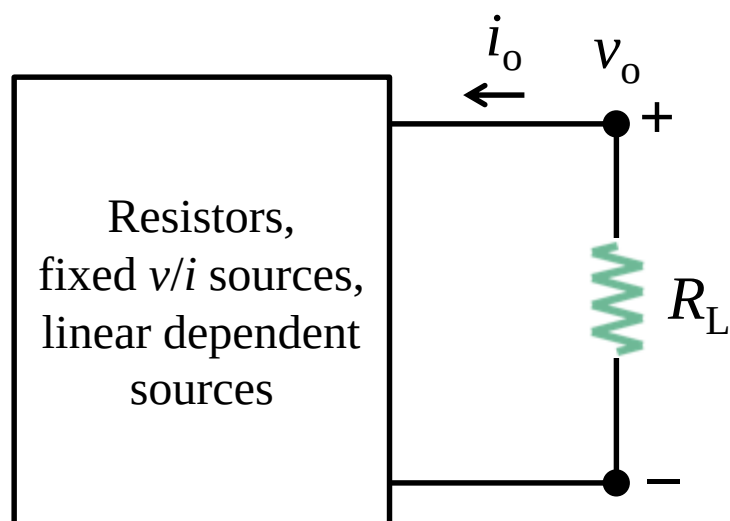
Short-circuit current: $i_{SC} = i_N = -i_o$

Equivalent R : $R_N = v_{OC}/i_{SC} = R_{eq}$

Thévenin-Norton transformation: $v_{Th} = i_N \cdot R_{Th}$

10.4 Thevenin's and Norton's theorems (d)

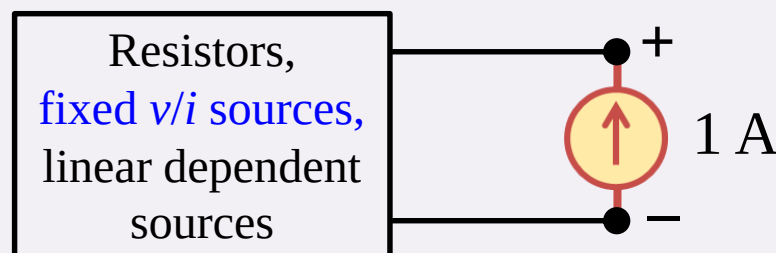
- Recall: to find v_{OC} , i_{SC} , R_{eq}



To find v_{OC} , remove the load and calculate the v_{OC} using nodal/mesh analysis.



To find i_{SC} , replace the load with an ideal wire and calculate the i_{SC} using nodal/mesh analysis.



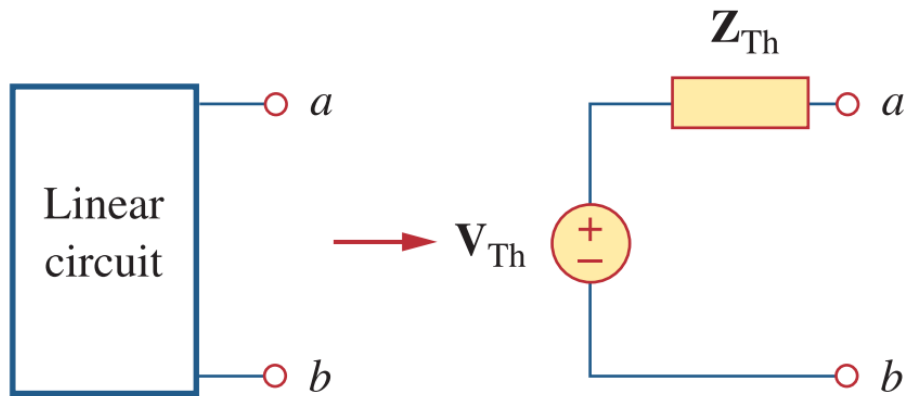
To find R_{eq} , replace the load with a 1 A current source ($i_O = 1\text{ A}$), turn off the independent sources, calculate the v_O , and $R_{eq} = v_O/i_O$.

*You can also use a 1 V voltage source

10.4 Thevenin's and Norton's theorems (e)

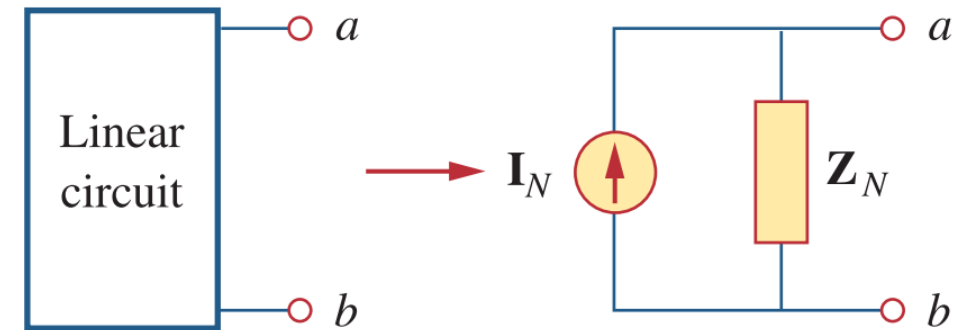
- Thevenin's and Norton's theorems for AC circuits

Thevenin equivalent



$$\begin{aligned} V_{Th} &= Z_N I_N \\ Z_{Th} &= Z_N \end{aligned}$$

Norton equivalent

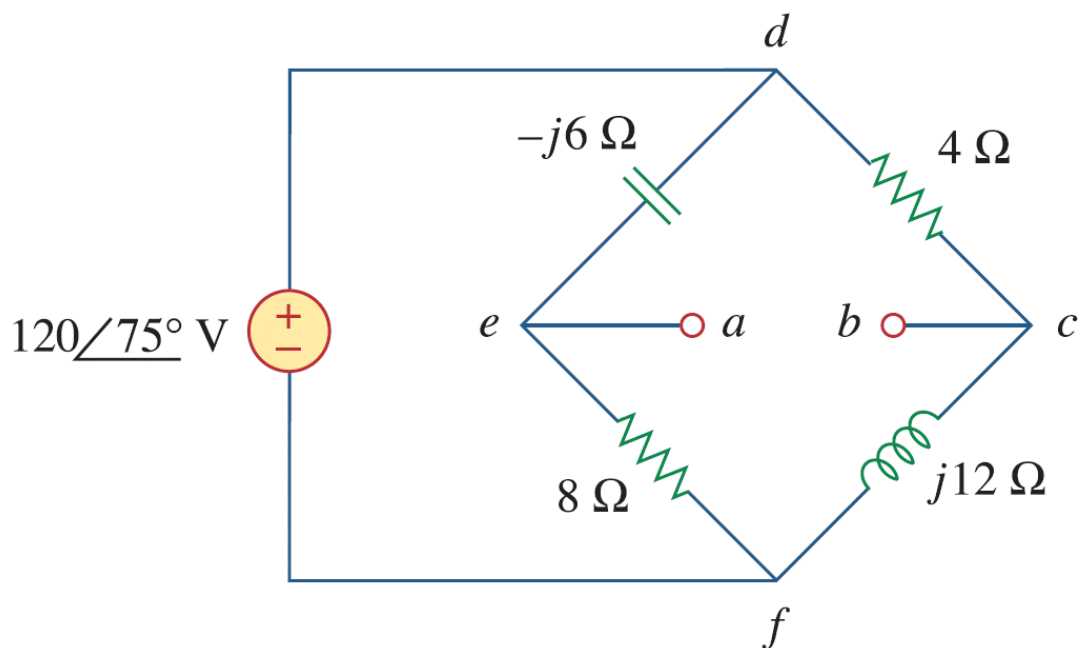


- a) Both Thevenin and Norton's theorems are applicable to AC circuits in phasor domain in the same way as DC. The only difference is the fact that the calculated values will be of complex number.

10.4 Thevenin's and Norton's theorems (f)

- Example 10.08

Example 10.08: Obtain the Thevenin equivalent at terminals a - b in the following circuits.



Solution:

(1) To find the V_{oc} :

$$\begin{aligned} V_{oc} &= \left(\frac{8}{8 - j6} - \frac{j12}{4 + j12} \right) V_i = \frac{j9 - 13}{50} V_i \\ &= -\frac{13 - j9}{50} V_i = j^2 \frac{13 - j9}{50} V_i = j^2 0.3162 \\ &= \mathbf{1\angle(90^\circ)} \mathbf{1\angle(90^\circ)} \mathbf{0.3162\angle(-34.69^\circ)} \mathbf{120\angle(75^\circ)} \\ &= 37.95 \angle(220.31^\circ) \text{ V} \end{aligned}$$

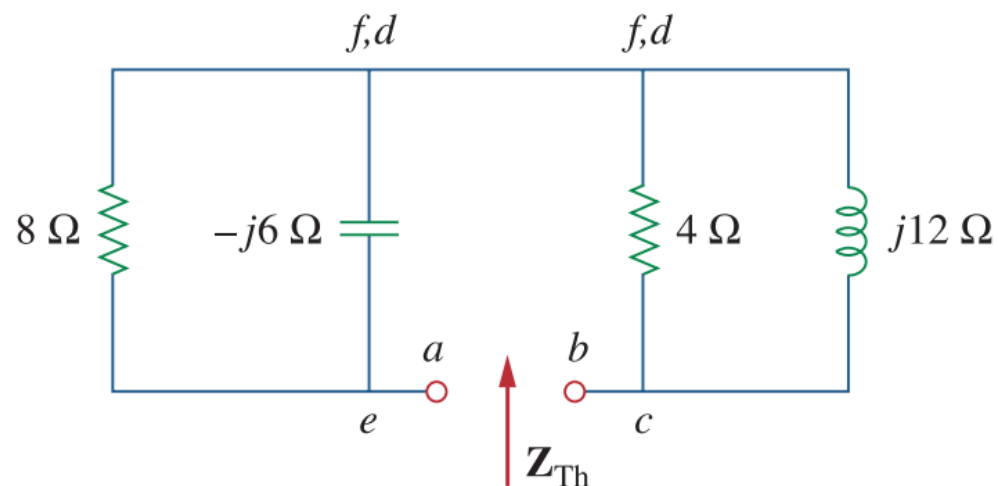
*Compare this to the solution on Page 427

10.4 Thevenin's and Norton's theorems (g)

- Example 10.08

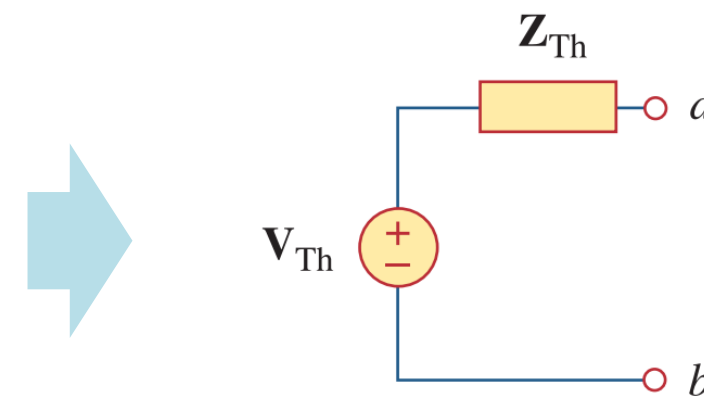
Example 10.08: Obtain the Thevenin equivalent at terminals a - b in the following circuits.

Solution: (2) To find the $\mathbf{Z}_{Th}$:



$$\mathbf{Z}_{Th} = \mathbf{Z}_1 + \mathbf{Z}_2 = 6.48 - j2.64 \, \Omega$$

(3) Then we have the Thevenin equivalent:



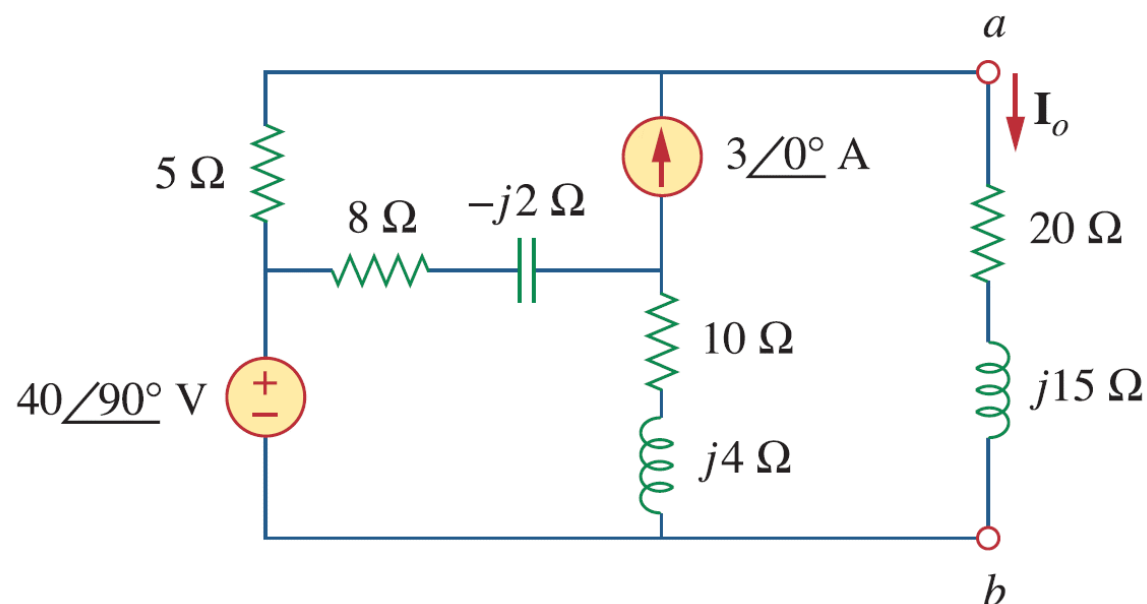
$$\mathbf{V}_{Th} = \mathbf{V}_{oc} = 37.95 \angle (220.31^\circ) \, \text{V}$$

$$\mathbf{Z}_{Th} = 6.48 + j2.64 \, \Omega$$

10.4 Thevenin's and Norton's theorems (h)

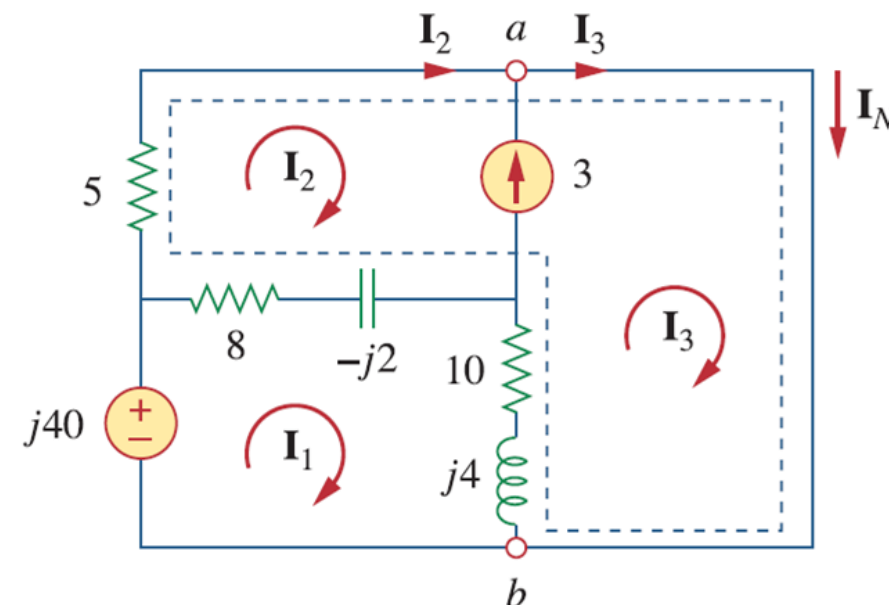
• Example 10.09

Example 10.09: Obtain current I_o using Norton's theorem.



Solution:

(1) To get I_{sc} :



For mesh 1:

$$-j40 + (18 + j2)I_1 - (8 - j2)I_2 - (10 + j4)I_3 = 0$$

For the supermesh:

$$(13 - j2)I_2 + (10 + j4)I_3 - (18 + j2)I_1 = 0$$

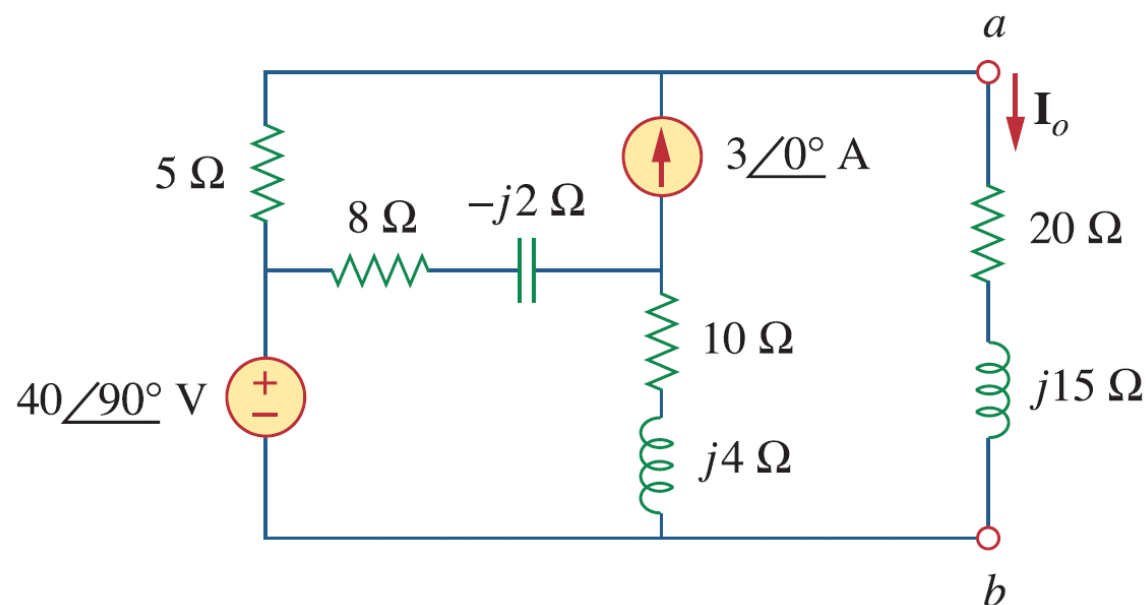
For the shared current source: $I_3 = I_2 + 3$

➡ $I_N = I_3 = (3 + j8) \text{ A}$

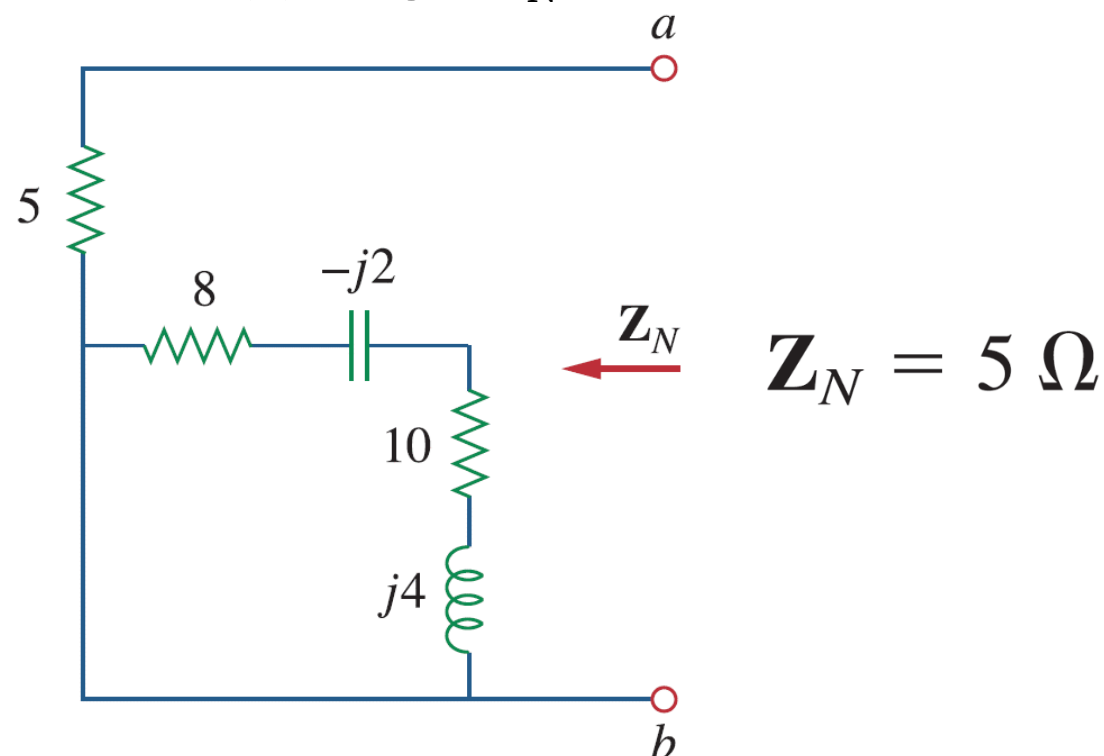
10.4 Thevenin's and Norton's theorems (i)

- Example 10.09

Example 10.09: Obtain current I_o using Norton's theorem.



Solution: (2) To get Z_N :

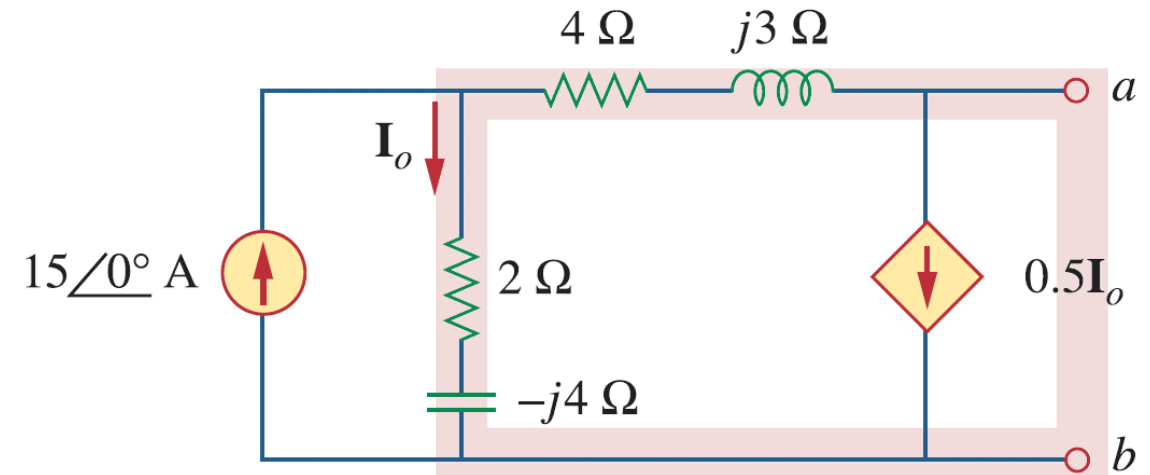


(3) Then you have the Norton equivalent based on I_{SC} and Z_N :

10.4 Thevenin's and Norton's theorems (j)

- Example 10.10

Example 10.10: Find the Thevenin equivalent of the following circuit as seen from terminals a - b .



Solution: (1) To find the V_{oc} :

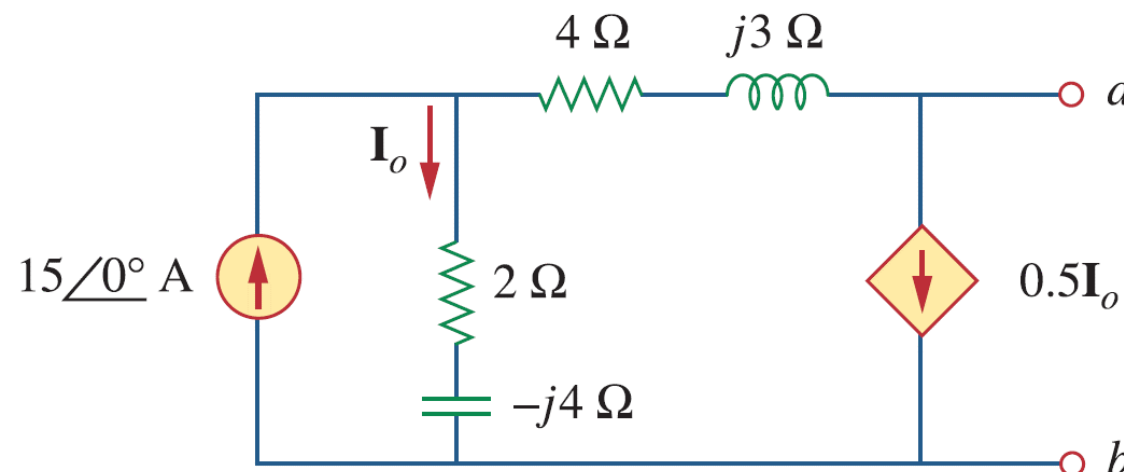
$$\text{KCL: } 15 = \mathbf{I}_o + 0.5\mathbf{I}_o \rightarrow \mathbf{I}_o = 10\angle 0^\circ \text{ A}$$

$$\text{KVL: } 0 = -\mathbf{I}_o(2 - j4) + 0.5\mathbf{I}_o(4 + j3) + \mathbf{V}_{oc} \rightarrow \mathbf{V}_{oc} = -j55 \text{ V} = 55\angle -90^\circ \text{ V}$$

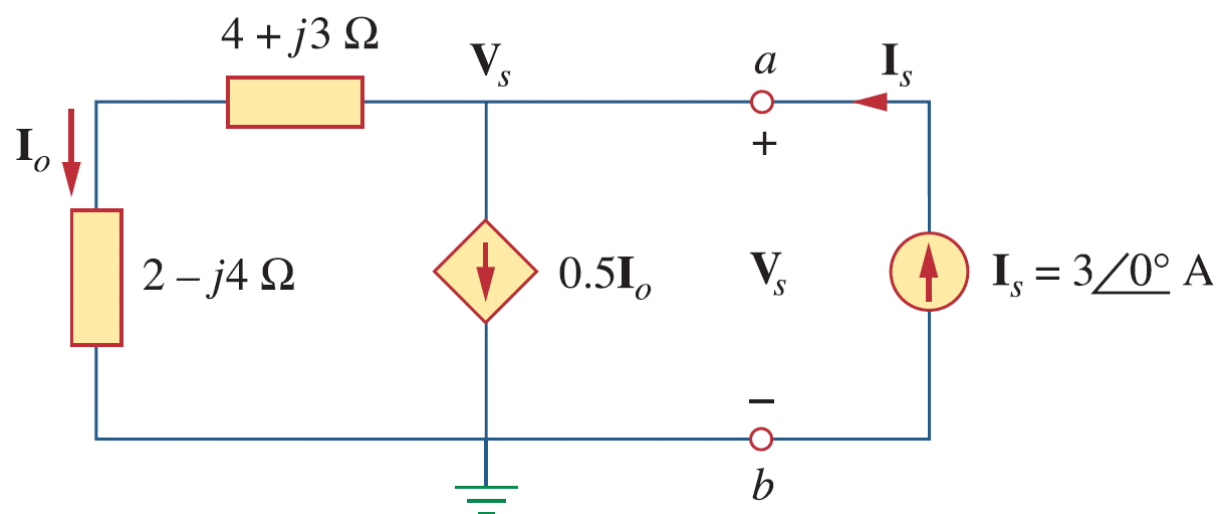
10.4 Thevenin's and Norton's theorems (k)

• Example 10.10

Example 10.10: Find the Thevenin equivalent of the following circuit as seen from terminals a - b .



Solution: (2) To find the $\mathbf{Z}_{Th}$:



$$\text{KCL: } 3 = \mathbf{I}_o + 0.5\mathbf{I}_o \rightarrow \mathbf{I}_o = 2\angle 0^\circ \text{ A}$$

$$\text{KVL: } \mathbf{V}_{ab} = \mathbf{I}_o (6 - j4 + j3) = 2(6 - j) \text{ V}$$

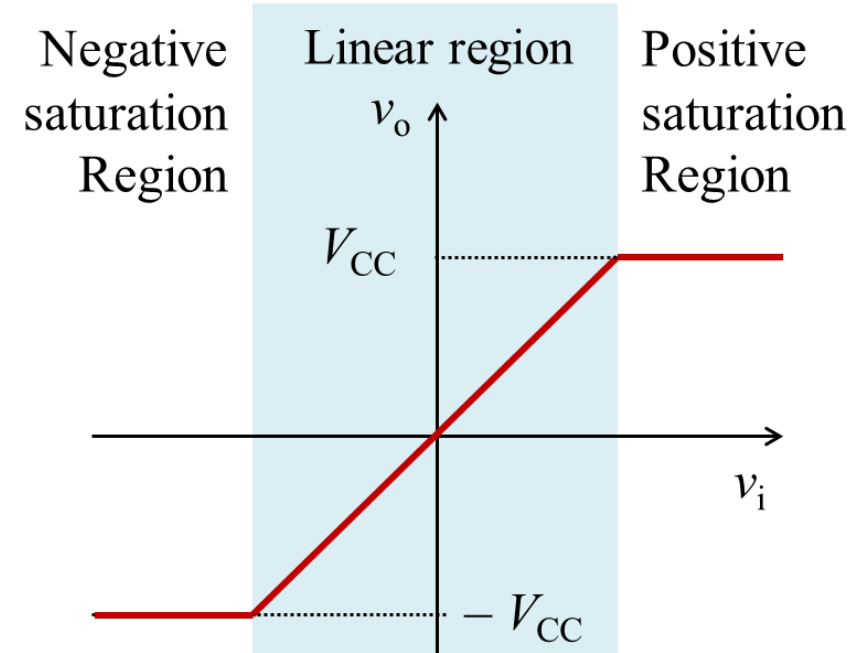
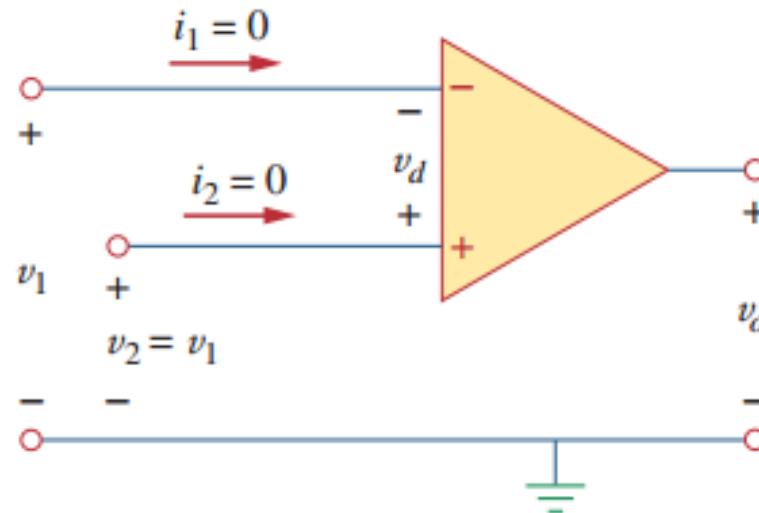
$$\mathbf{Z}_{Th} = \frac{\mathbf{V}_s}{\mathbf{I}_s} = \frac{2(6 - j)}{3} = 4 - j0.6667 \Omega$$

Lesson 10 Steady-state and power analysis of AC circuits

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10.4	Thevenin and Norton equivalent circuits	Slides 42-52
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10.5 Thevenin's and Norton's theorems (a)

- Recall: circuits with ideal op-amp



Golden rules for analyzing ideal op-amp:

- (1) Currents into both input terminals are 0: $i_1 = i_2 = 0$ (虚断)
- (2) The voltage across the input terminals is 0: $v_2 - v_1 = 0$ (虚短)

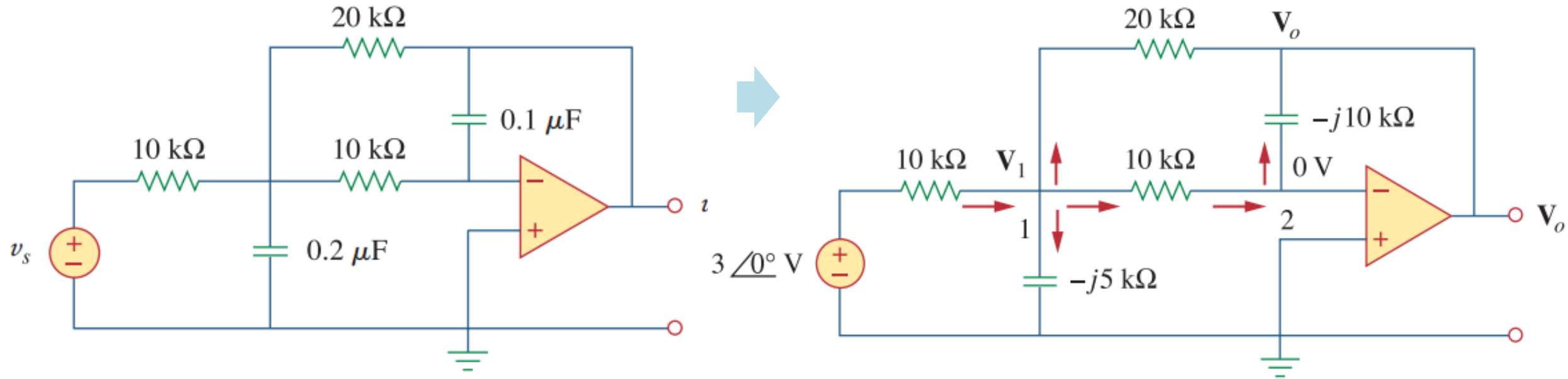
For ac circuits, phasors can be used for op-amp as long as it operates **in linear region**

10.5 Thevenin's and Norton's theorems (b)

- Example 10.11

Example 10.11: Determine $v_o(t)$ for the op amp circuit if $v_s = 3 \cos(1000t)$ V.

Solution: (1) Transform the circuit to phasor domain

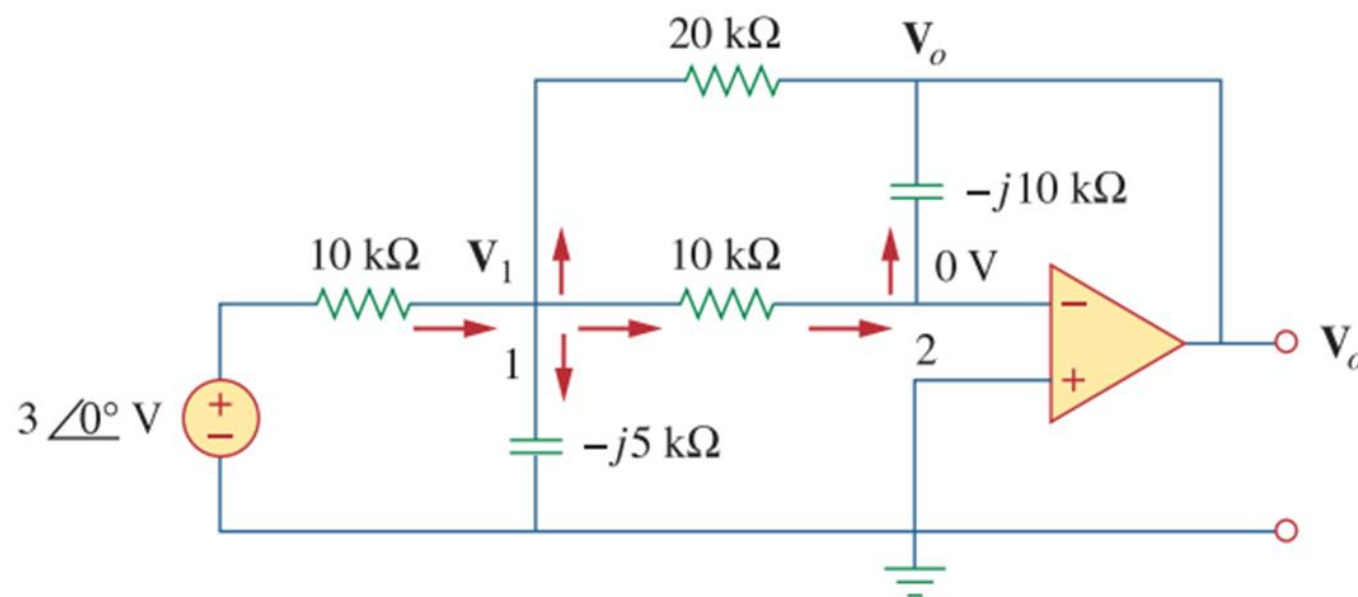


10.5 Thevenin's and Norton's theorems (c)

• Example 10.11

Example 10.11: Determine $v_o(t)$ for the op amp circuit if $v_s = 3 \cos(1000t)$ V.

Solution: (2) Base on nodal analysis, use nodal analysis.



For node 1:

$$\frac{3\angle 0^\circ - V_1}{10} = \frac{V_1}{-j5} + \frac{V_1 - 0}{10} + \frac{V_1 - V_o}{20}$$

For node 2:
$$\frac{V_1 - 0}{10} = \frac{0 - V_o}{-j10}$$

$$6 = (5 + j4)V_1 - V_o \quad \& \quad V_1 = -jV_o$$

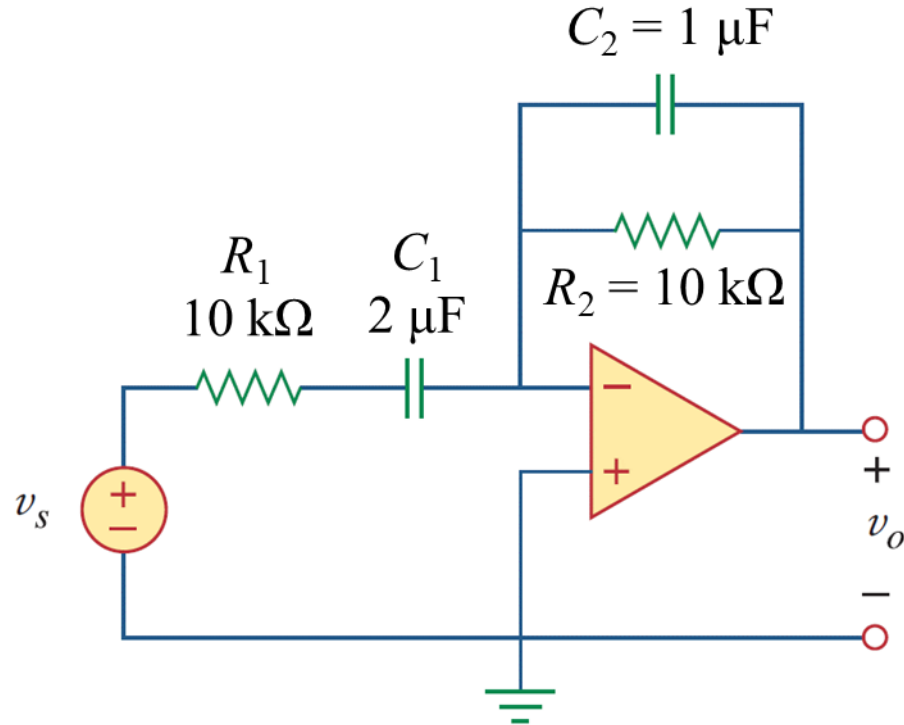
$$V_o = \frac{6}{3 - j5} = 1.029\angle 59.04^\circ$$

$$v_o(t) = 1.029 \cos(1000t + 59.04^\circ) \text{ V}$$

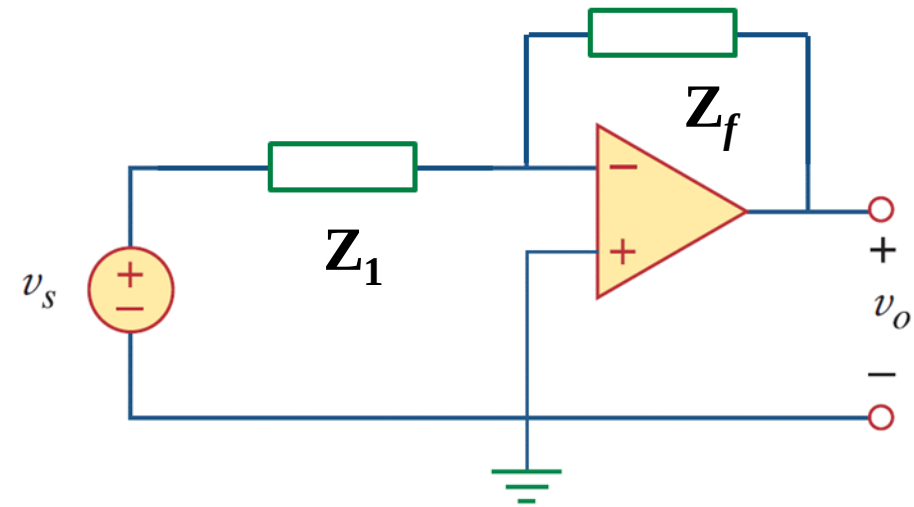
10.5 Thevenin's and Norton's theorems (d)

- Example 10.12

Example 10.12: Compute the closed-loop gain and the phase shift in the following circuit.



Solution: Simplify the circuits

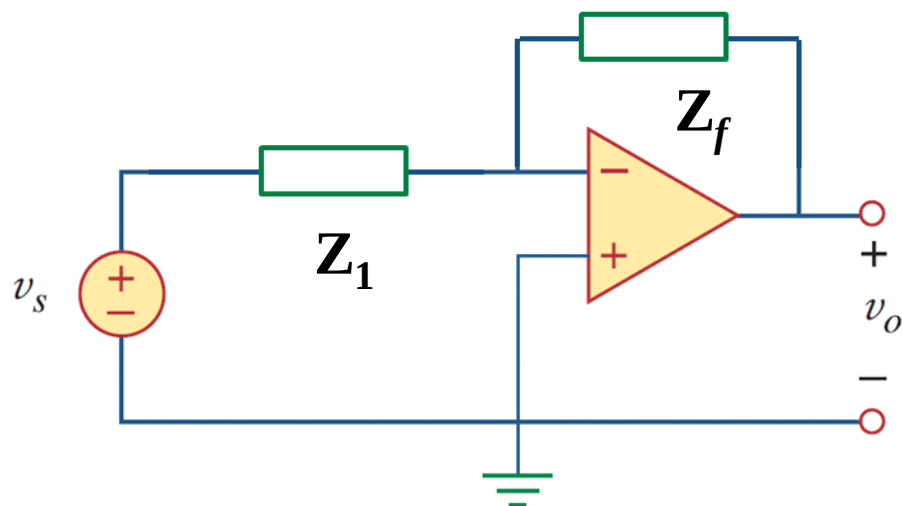


10.5 Thevenin's and Norton's theorems (e)

- Example 10.12

Example 10.12: Compute the closed-loop gain and the phase shift in the following circuit.

Solution:



$$\begin{aligned} Z_f &= R_2 \parallel \frac{1}{j\omega C_2} = \frac{R_2}{1 + j\omega R_2 C_2} \\ Z_i &= R_1 + \frac{1}{j\omega C_1} = \frac{1 + j\omega R_1 C_1}{j\omega C_1} \\ \mathbf{G} &= \frac{\mathbf{V}_o}{\mathbf{V}_s} = -\frac{\mathbf{Z}_f}{\mathbf{Z}_i} = \frac{-j\omega C_1 R_2}{(1 + j\omega R_1 C_1)(1 + j\omega R_2 C_2)} \\ &= \frac{-j4}{(1 + j4)(1 + j2)} = 0.434 \angle 130.6^\circ \end{aligned}$$

Lesson 10 Steady-state and power analysis of AC circuits

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10.6 AC power in time domain (a)

- Recall: power

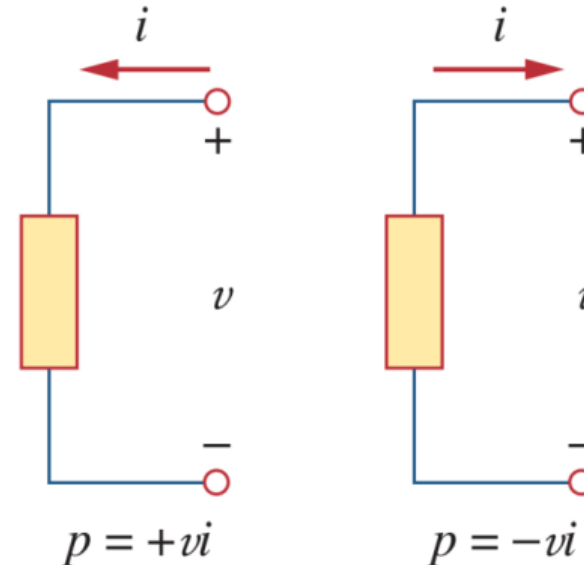
a) Power is defined as the rate of energy provided or consumed by a device per unit time. It is measured in Watt (W).

$$p \triangleq \frac{dw}{dt}$$

$$w = \int_{t_0}^t p dt = \int_{t_0}^t vi dt$$

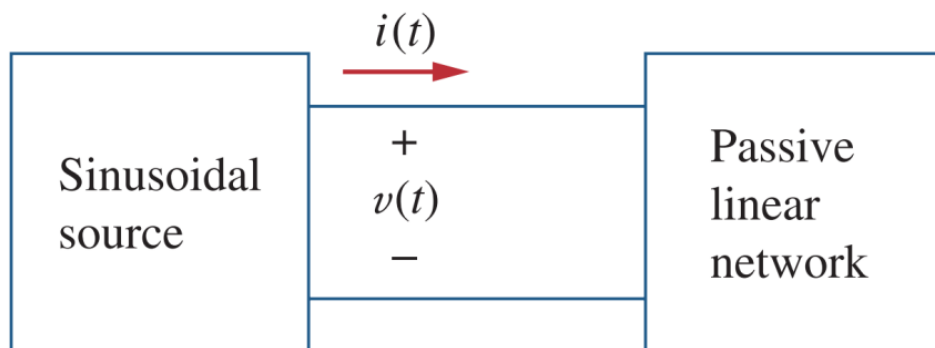
b) Power is the product of voltage and current:

$$p = vi$$



10.6 AC power in time domain (b)

- Instantaneous Power (瞬时功率)



$$v(t) = V_m \cos(\omega t + \theta_v)$$

$$i(t) = I_m \cos(\omega t + \theta_i)$$

$$p(t) = v(t)i(t) = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

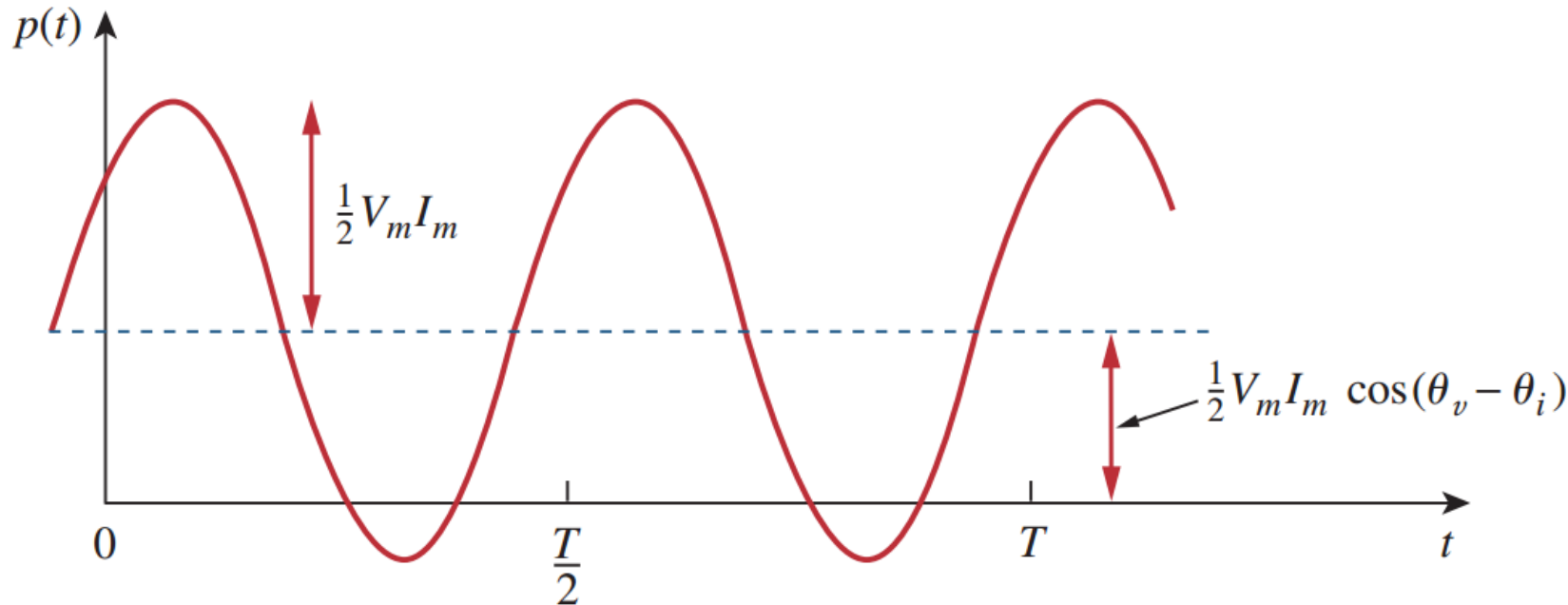
$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

10.6 AC power in time domain (c)

- Instantaneous Power (瞬时功率)

$$p(t) = \boxed{\frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)} + \boxed{\frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)}$$



The power is consist of two components.

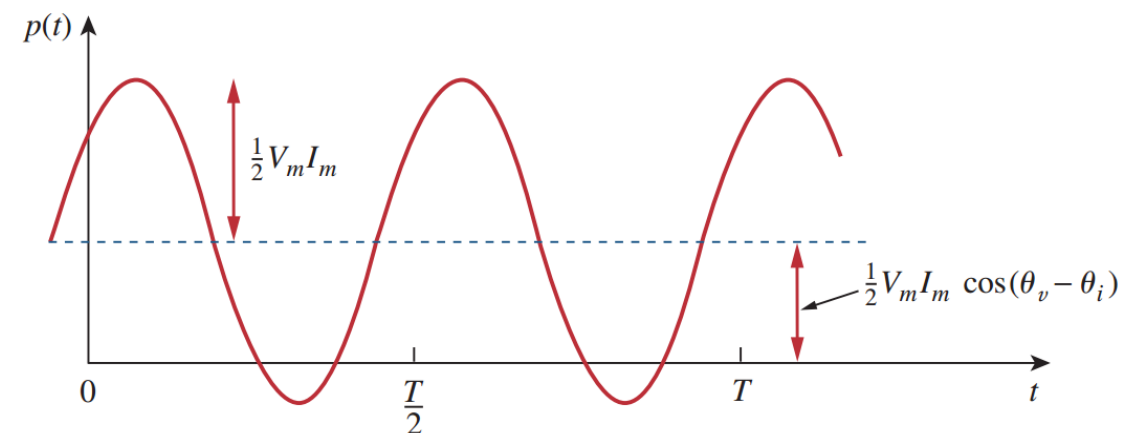
(1) A constant dependent on the phase difference between the voltage and current

(2) A sinusoid oscillating at twice the frequency of the voltage/ current.

10.6 AC power in time domain (d)

- Average Power (平均功率)

(a) The average power, in watts, is the average of the instantaneous power over one period (or a very long time $t \rightarrow \infty$).

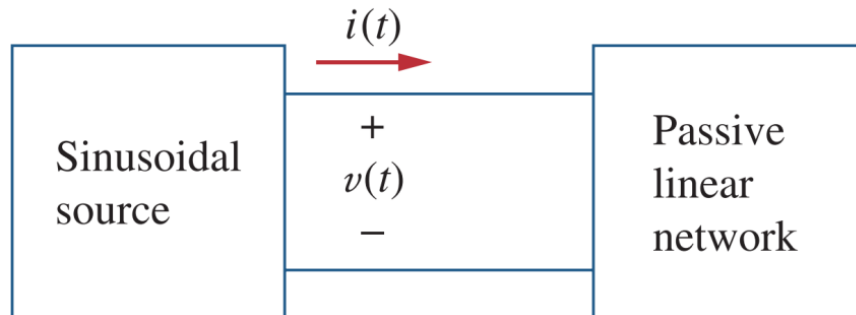


$$\begin{aligned}
 P &= \frac{1}{T} \int_0^T p(t) dt = \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) dt + \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i) dt \\
 &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \frac{1}{T} \int_0^T dt + \frac{1}{2} V_m I_m \frac{1}{T} \int_0^T \cos(2\omega t + \theta_v + \theta_i) dt \\
 &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2} \text{Re}[V_m \angle \theta_v I_m \angle (-\theta_i)] = \frac{1}{2} \text{Re}[\mathbf{VI}^*]
 \end{aligned}$$

10.6 AC power in time domain (e)

- Average Power (平均功率)

(a) A resistive load (R) absorbs power at all time, while a reactive load (L or C) absorbs zero average power.



$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

When $\theta_v = \theta_i$, resistive load:

$$P = \frac{1}{2} V_m I_m = \frac{1}{2} I_m^2 R = \frac{1}{2} |\mathbf{I}|^2 R$$

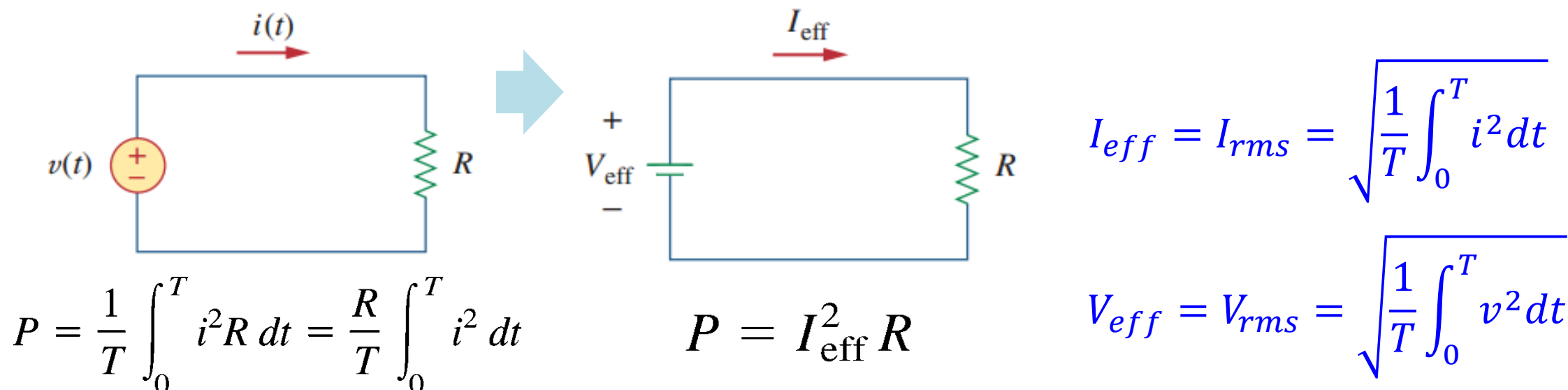
When $\theta_v - \theta_i = \pm 90^\circ$, reactive load:

$$P = \frac{1}{2} V_m I_m \cos 90^\circ = 0$$

10.6 AC power in time domain (f)

- Effective value/RMS value (有效值/均方根值)

- (a) The idea of **effective value** arises from the need to measure the effectiveness of a voltage or current source in delivering power to a resistive load.
- (b) The effective value of **a periodic current** is the dc current that delivers **the same average power to a resistor** as the periodic current.
- (c) The effective value of a periodic signal is its root mean square (RMS) value.


$$P = \frac{1}{T} \int_0^T i^2 R dt = \frac{R}{T} \int_0^T i^2 dt$$
$$P = I_{\text{eff}}^2 R$$
$$I_{\text{eff}} = I_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$
$$V_{\text{eff}} = V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T v^2 dt}$$

10.6 AC power in time domain (g)

- RMS values of sinusoids:

$$I_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T I_m^2 \cos^2 \omega t \, dt}$$
$$= \sqrt{\frac{I_m^2}{T} \int_0^T \frac{1}{2} (1 + \cos 2\omega t) \, dt}$$



$$I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$$

And similarly, $V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$



(a) The average power can be written as:

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Or:
$$P = I_{\text{rms}}^2 R \cos(\theta_v - \theta_i) = \frac{v_{\text{rms}}^2}{R} \cos(\theta_v - \theta_i)$$

10.6 AC power in time domain (h)

- The apparent power and the power factor:

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Apparent power (视在功率)

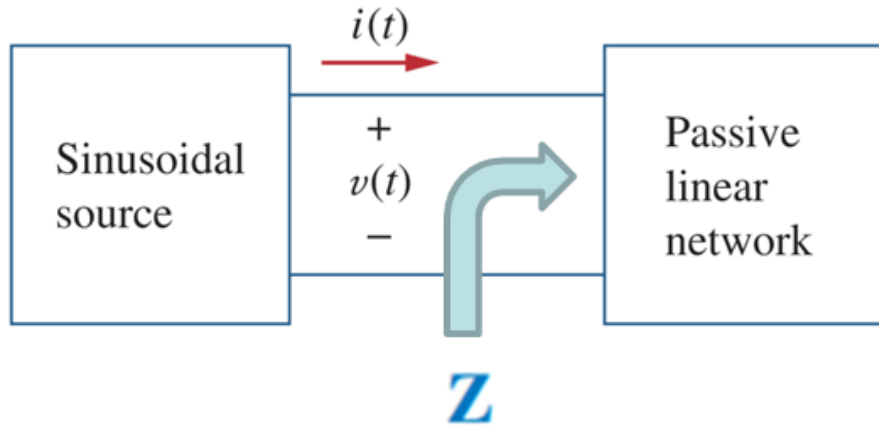
- The amplitude is called the apparent power, and denoted as S .
- $S = V_{\text{rms}} I_{\text{rms}}$
- measured in VA

Power factor (pf, 功率因数)

- $pf = P/S = \cos(\theta_v - \theta_i)$
- The ratio between the P to the S is called power factor (pf).
- Therein, the angle $\theta_v - \theta_i$ is called the power factor angle.

10.6 AC power in time domain (i)

- The power factor:



$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{V_m \angle \theta_v}{I_m \angle \theta_i} = \frac{V_m}{I_m} \angle \theta_v - \theta_i = \frac{V_{\text{rms}}}{I_{\text{rms}}} \angle \theta_v - \theta_i$$

- The power factor is **the cosine of the phase difference** between voltage and current. It is also **the cosine of the angle of the load impedance**.
- The power factor takes value in the range of $[0, 1]$.
- $pf = 1$ for purely resistive loads; $pf = 0$ for purely reactive (inductive or capacitive) loads.
- The power factor is leading if the current leads the voltage.

10.6 AC power in time domain (j)

- Brief summary :

Instantaneous power
瞬时功率

$$p(t) = \boxed{\frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)} + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

Average power
平均功率

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Effective values
有效值

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}}, \quad I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$$

Power factor | 功率因数

$$\text{pf} = P/S = \cos(\theta_v - \theta_i)$$

Apparent power | 视在功率

$$S = V_{\text{rms}} I_{\text{rms}}$$

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{\mathbf{V}_{\text{rms}}}{\mathbf{I}_{\text{rms}}} = \frac{V_{\text{rms}}}{I_{\text{rms}}} \angle \theta_v - \theta_i$$

10.6 AC power in time domain (k)

- Example 10.13:

Example 10.13: A series-connected load draws a current $i(t) = 4 \cos(100\pi t + 10^\circ)$ A when the applied voltage is $v(t) = 120 \cos(100\pi t - 20^\circ)$ V. Find the apparent power and the power factor of the load. Determine the element values that form the series-connected load.

Solution: The apparent power: $S = V_{\text{rms}} I_{\text{rms}} = \frac{120}{\sqrt{2}} \frac{4}{\sqrt{2}} = 240 \text{ VA}$

The power factor: $\text{pf} = \cos(\theta_v - \theta_i) = \cos(-20^\circ - 10^\circ) = 0.866$ *i leads v
leading pf*

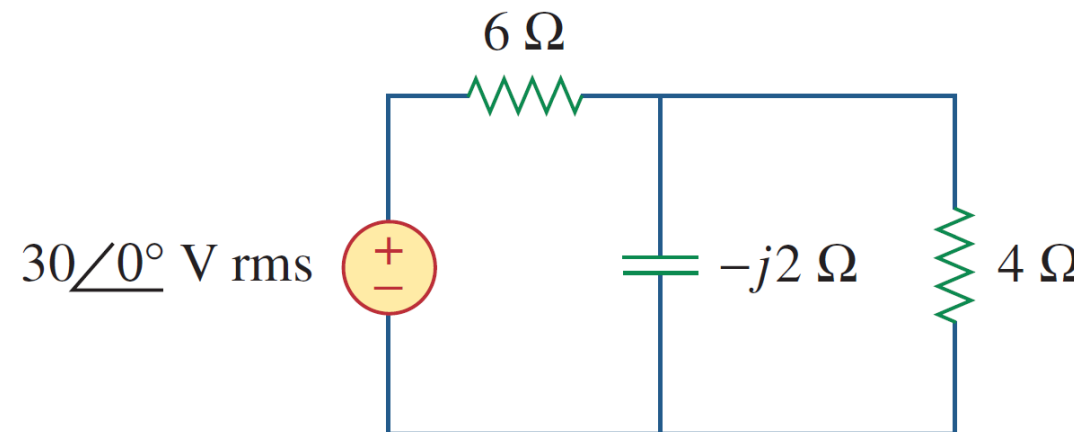
$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{120 \angle -20^\circ}{4 \angle 10^\circ} = 30 \angle -30^\circ = 25.98 - j15 \Omega \quad X_C = -15 = -\frac{1}{\omega C}$$

A 25.98 Ω resistors in series with a 212.2 μF capacitor

10.6 AC power in time domain (I)

- Example 10.14:

Example 10.14: (a) Determine the power factor of the entire circuit as seen by the source. (b) Calculate the average power delivered by the source.



Solution:
$$\mathbf{Z} = 6 + 4 \parallel (-j2) = 6 + \frac{-j2 \times 4}{4 - j2} = 6.8 - j1.6 = 7 \angle -13.24^\circ \Omega$$

$$\text{pf} = \cos(-13.24) = 0.9734 \quad (i \text{ leads } v \mid \text{leading pf})$$

$$\mathbf{I}_{\text{rms}} = \frac{\mathbf{V}_{\text{rms}}}{\mathbf{Z}} = \frac{30 \angle 0^\circ}{7 \angle -13.24^\circ} = 4.286 \angle 13.24^\circ \text{ A}$$

$$P = V_{\text{rms}} I_{\text{rms}} \text{pf} = (30)(4.286)0.9734 = 125 \text{ W}$$

Lesson 10 Steady-state and power analysis of AC circuits

10.1	Nodal and mesh analyses	Slides 13-24
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10.7 AC power in phasor domain (a)

- Complex power (复功率)

In time domain: $v(t) = V_m \cos(\omega t + \theta_v)$ and $i(t) = I_m \cos(\omega t + \theta_i)$

In Phasor domain: $\mathbf{V} = V_m \angle \theta_v$ and $\mathbf{I} = I_m \angle \theta_i$

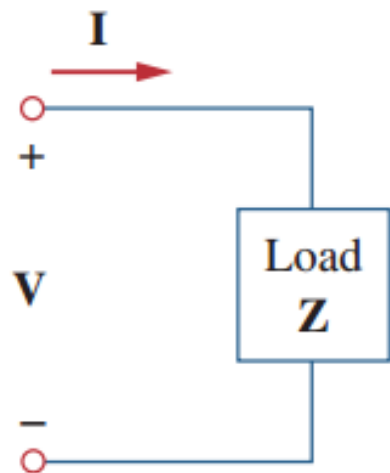
a) In analogy to voltage and current, we define a new phasor for power – **complex power**

$$S = \frac{1}{2} \mathbf{V} \mathbf{I}^* = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^*$$

Where, $\mathbf{V}_{\text{rms}} = \frac{\mathbf{V}}{\sqrt{2}} = V_{\text{rms}} \angle \theta_v$, $\mathbf{I}_{\text{rms}} = \frac{\mathbf{I}}{\sqrt{2}} = I_{\text{rms}} \angle \theta_i$ are **effective voltage and current phasors**.

10.7 AC power in phasor domain (b)

- Real power (有功功率) and reactive power (无功功率)



$$\mathbf{S} = V_{\text{rms}} I_{\text{rms}} \angle \theta_v - \theta_i$$

$$= V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) + j V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i)$$

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{V_{\text{rms}}}{I_{\text{rms}}} \angle \theta_v - \theta_i \quad \mathbf{S} = I_{\text{rms}}^2 \mathbf{Z} = \frac{V_{\text{rms}}^2}{\mathbf{Z}^*} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^*$$

Since $\mathbf{Z} = R + jX$, $\mathbf{S} = I_{\text{rms}}^2 R + j I_{\text{rms}}^2 X = P + jQ$

P is the average or the real power: $P = \text{Re}[\mathbf{S}] = I_{\text{rms}}^2 R = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$

Q is the reactive power: $Q = \text{Im}[\mathbf{S}] = I_{\text{rms}}^2 X = V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i)$

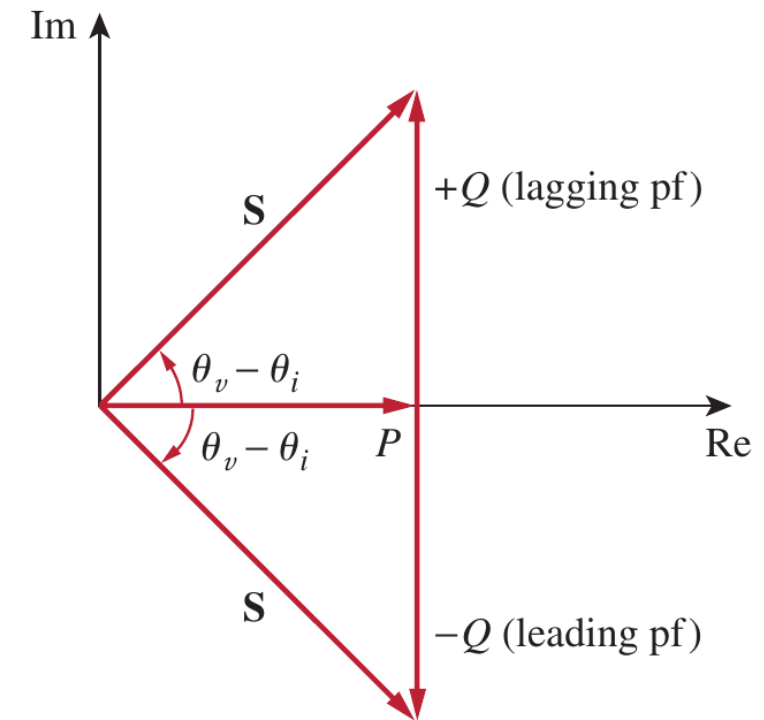
10.7 AC power in phasor domain (c)

- Real power (有功功率) and reactive power (无功功率)

a) Complex power, measured in **VA**, is the product of the rms voltage phasor and the complex conjugate of the rms current phasor. As a complex quantity, its real part is real power P and its imaginary part is reactive power Q .

- $Q = 0$ for resistive loads (unity pf).
- $Q < 0$ for capacitive loads (leading pf).
- $Q > 0$ for inductive loads (lagging pf).

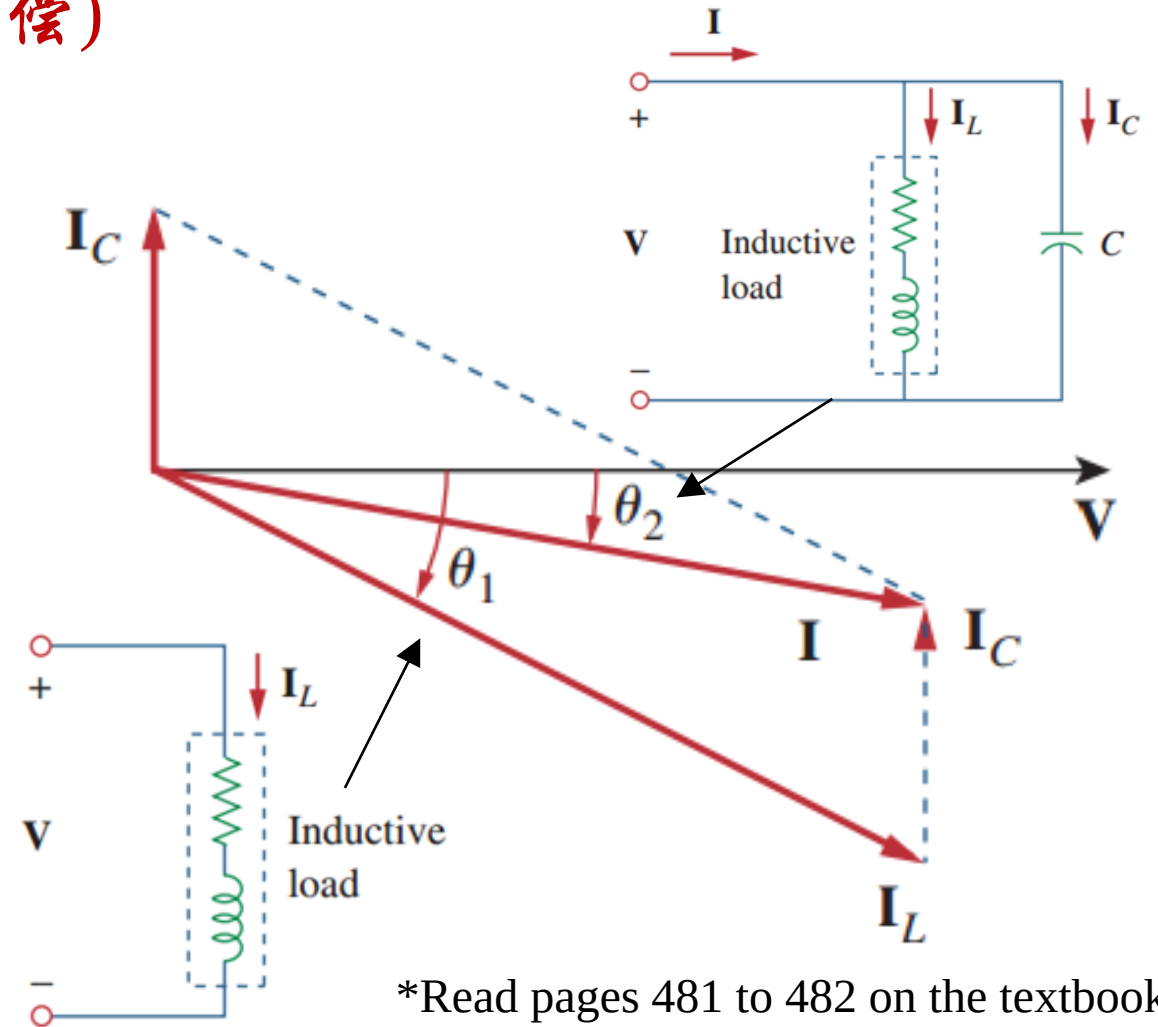
- b) The real power P is the only useful power. It is measured in **watts**.
- c) The reactive power Q is a measure of the energy exchange between the source and the reactive load. It is measured in units of **volt-ampere reactive (VAR, 乏)**



10.7 AC power in phasor domain (d)

- Power factor correction (无功补偿)

- a) Drawback of low power factor: energy exchange between the source and the load that has no contribution to actual power transfer. For the same mount of energy transfer, it requires larger current (high bills), which makes the ‘power supply system’ hard to operate.
- b) The process of increasing the power factor without altering the voltage and or current to the original load is known as power factor correction.

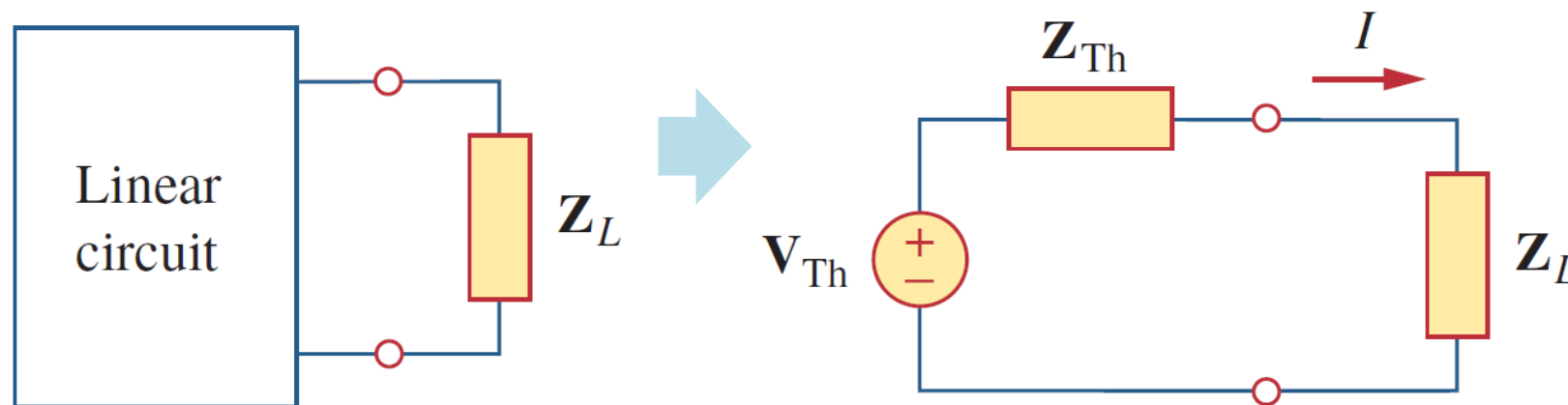


*Read pages 481 to 482 on the textbook

10.7 AC power in phasor domain (e)

- Maximum average power transfer (最大平均功率传输)

(a) For DC circuits, we have proved that the maximum power is transferred to the load when $R_L = R_{Th}$. This result can be extended to ac circuits for maximum average power transfer.



$$Z_L = R_L + jX_L = R_{Th} - jX_{Th} = Z_{Th}^*$$

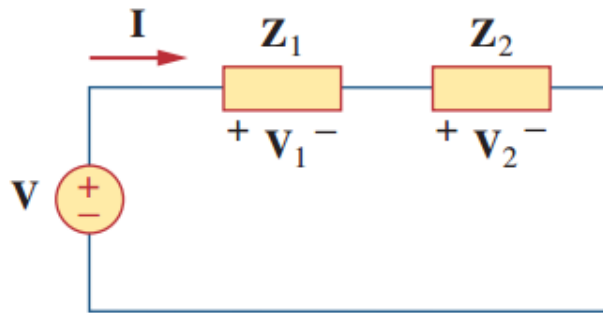
*Read pages 464 to 465 on the textbook

$$P_{\max} = \frac{|V_{Th}|^2}{8R_{Th}}$$

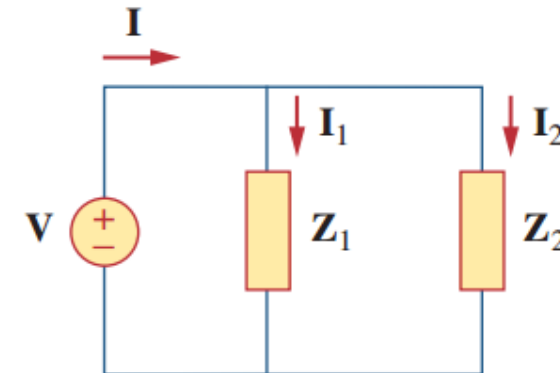
10.7 AC power in phasor domain (f)

- Conservation of AC power

- The principle of conservation of power applies to ac circuits same as to dc circuits.
- The complex, real, and reactive powers of the sources equal the respective sums of the complex, real and reactive powers of the individual loads.



$$\mathbf{S} = \mathbf{V}\mathbf{I}^* = (\mathbf{V}_1 + \mathbf{V}_2)\mathbf{I}^* = \mathbf{V}_1\mathbf{I}^* + \mathbf{V}_2\mathbf{I}^* = \mathbf{S}_1 + \mathbf{S}_2$$



$$\mathbf{S} = \mathbf{V}\mathbf{I}^* = \mathbf{V}(\mathbf{I}_1^* + \mathbf{I}_2^*) = \mathbf{V}\mathbf{I}_1^* + \mathbf{V}\mathbf{I}_2^* = \mathbf{S}_1 + \mathbf{S}_2$$

10.7 AC power in phasor domain (g)

• Example 10.15:

Example 10.15: The voltage across a load is $v(t) = 60 \cos(\omega t - 10^\circ) \text{ V}$ and the current through the element in the direction of the voltage drop is $i(t) = 1.5 \cos(\omega t + 50^\circ) \text{ A}$. Find: a) the complex and apparent powers, b) the real and reactive powers and c) the power factor and the load impedance.

Solution :

(a) To find the complex and apparent powers:

Find the rms phasors for voltage and current

$$\mathbf{V}_{\text{rms}} = \frac{60}{\sqrt{2}} \angle -10^\circ, \quad \mathbf{I}_{\text{rms}} = \frac{1.5}{\sqrt{2}} \angle +50^\circ$$

The complex power is

$$\mathbf{S} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^* = \left(\frac{60}{\sqrt{2}} \angle -10^\circ \right) \left(\frac{1.5}{\sqrt{2}} \angle -50^\circ \right) = 45 \angle -60^\circ \text{ VA}$$

So the apparent power is 45 VA

10.7 AC power in phasor domain (h)

- Example 10.15:

Example 10.15: The voltage across a load is $v(t) = 60 \cos(\omega t - 10^\circ) \text{ V}$ and the current through the element in the direction of the voltage drop is $i(t) = 1.5 \cos(\omega t + 50^\circ) \text{ A}$. Find: a) the complex and apparent powers, b) the real and reactive powers and c) the power factor and the load impedance.

Solution :

(b) To find the real and reactive powers:

Rewrite the complex power in rectangular form

$$\begin{aligned} S &= 45 \angle (-60^\circ) \\ &= 45 [\cos(-60^\circ) + j \sin(-60^\circ)] \\ &= 22.5 - j38.97 \text{ VA} \end{aligned}$$

Since $S = P + jQ$, the real power is $P = 22.5 \text{ W}$; while the reactive power is $Q = -38.97 \text{ VAR}$;

10.7 AC power in phasor domain (i)

• Example 10.15:

Example 10.15: The voltage across a load is $v(t) = 60 \cos(\omega t - 10^\circ) \text{ V}$ and the current through the element in the direction of the voltage drop is $i(t) = 1.5 \cos(\omega t + 50^\circ) \text{ A}$. Find: a) the complex and apparent powers, b) the real and reactive powers and c) the power factor and the load impedance.

Solution :

(c) To find the power factor and the load impedance:

According to the complex power:

$$\mathbf{S} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^* = 45 \angle (-60^\circ) \text{ VA}$$

So the power factor is: $\text{pf} = \cos(-60^\circ) = 0.5$ (*leading*)

The load impedance:

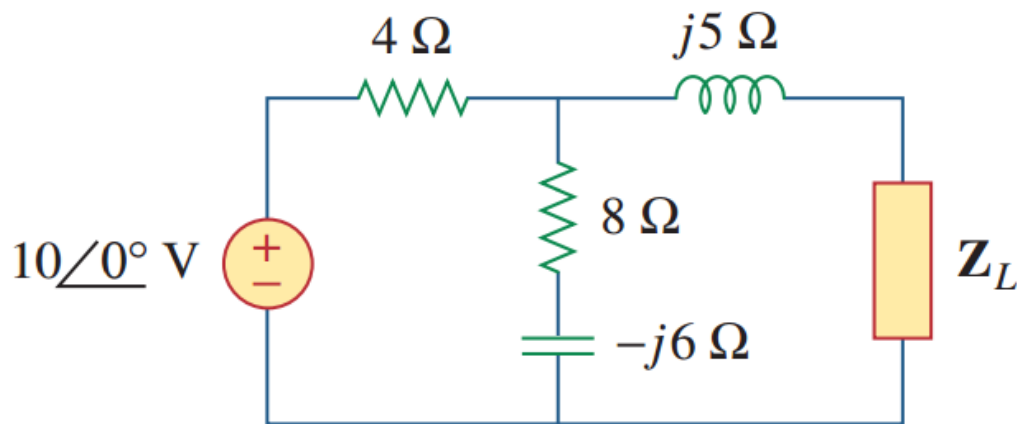
$$\mathbf{Z} = \frac{\mathbf{V}_{\text{rms}}}{\mathbf{I}_{\text{rms}}} = \frac{60 \angle (-10^\circ)}{1.5 \angle (+50^\circ)} = 40 \angle (-60^\circ) \Omega$$

This is a capacitive impedance.

10.7 AC power in phasor domain (i)

• Example 10.16:

Example 10.16: Determine the load impedance $\mathbf{Z}_L$ that maximizes the average power drawn from the circuit. What is the maximum average power?



Solution :

(1) Find the Thevenin equivalent:

$$\mathbf{V}_{\text{Th}} = \frac{8 - j6}{4 + 8 - j6}(10) = 7.454 \angle -10.3^\circ \text{ V}$$

$$\begin{aligned} \mathbf{Z}_{\text{Th}} &= j5 + 4 \parallel (8 - j6) = j5 + \frac{4(8 - j6)}{4 + 8 - j6} \\ &= 2.933 + j4.467 \, \Omega \end{aligned}$$

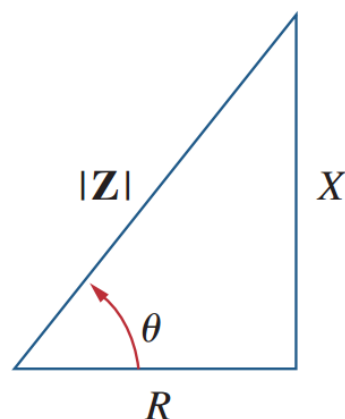
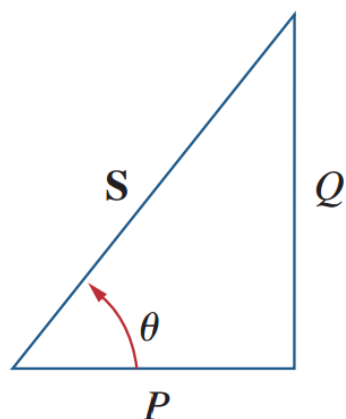
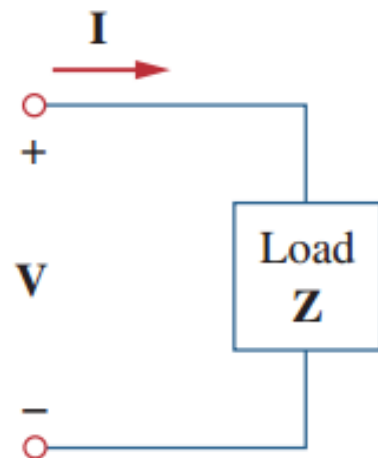
(2) Use the maximum power transfer theorem:

$$\mathbf{Z}_L = \mathbf{Z}_{\text{Th}}^* = 2.933 - j4.467 \, \Omega$$

$$P_{\text{max}} = \frac{|\mathbf{V}_{\text{Th}}|^2}{8R_{\text{Th}}} = \frac{(7.454)^2}{8(2.933)} = 2.368 \text{ W}$$

10.7 AC power in phasor domain (j)

- Brief summary:



$$\text{Complex Power} = S = P + jQ = \mathbf{V}_{\text{rms}}(\mathbf{I}_{\text{rms}})^*$$

$$= |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| \angle \theta_v - \theta_i$$

$$\text{Apparent Power} = S = |\mathbf{S}| = |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}| = \sqrt{P^2 + Q^2}$$

$$\text{Real Power} = P = \text{Re}(\mathbf{S}) = S \cos(\theta_v - \theta_i)$$

$$\text{Reactive Power} = Q = \text{Im}(\mathbf{S}) = S \sin(\theta_v - \theta_i)$$

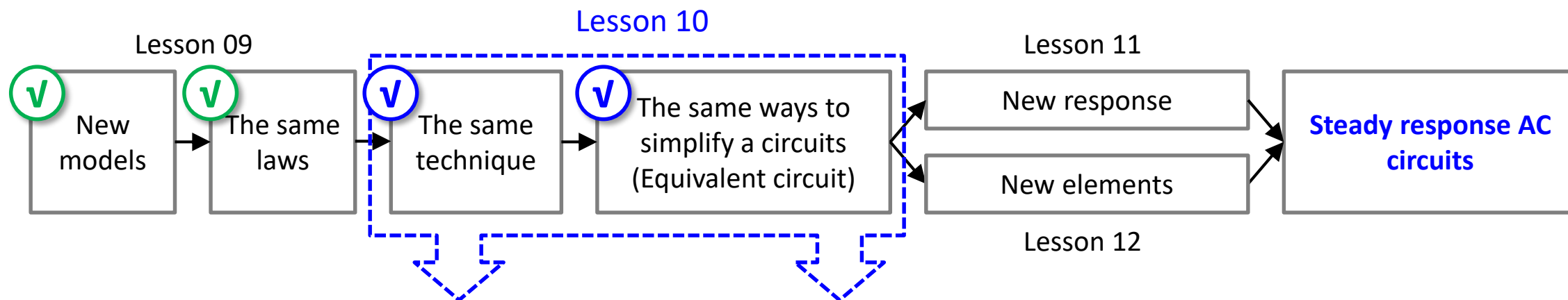
$$\text{Power Factor} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

Lesson 10 Steady-state and power analysis of AC circuits

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10.8 Summary and Assignment (a)

- EE104 road map



Techniques and theorems

- Nodal analysis and mesh analysis
- Superposition and source transformation
- Thevenin and Norton theorems
- AC op amp circuits

Power

- Power in time domain
- Power in phasor domain (complex power)
- Power factor
- Maximum power transfer theorem

10.8 Summary and Assignment (b)

Problems

10.7, 10.28, 10.66, 10.68, 11.6, 11.41, 11.53, 11.95

- (a) A certain amount of practicing is necessary to sharpen your mind and to deepen your understanding.
- (b) There are plenty of problems in the textbook, try to complete as many as you can.

Asperatus clouds:

